Information and Preferences in Shareholder Voting*

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Abstract

We develop a structural model of shareholder voting with incomplete information about proposal quality. We show that preferences and information are not separately identified from individual voting patterns alone: correlated information and strategic voting jointly determine the mapping from preferences to votes. Identification arises through cross-shareholder voting correlations and differential exposure to public information across investor types. Estimating the model on mutual fund voting, we recover institution-level information precision and information-corrected preference parameters. Despite higher support for management, blockholders' preferences are close to those of dispersed shareholders. Equalizing information across investor types has a larger effect on voting than equalizing preferences.

Keywords: corporate governance, shareholder voting, institutional investors, strategic voting

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1. Introduction

Institutional investors occupy blockholder positions in practically all listed firms in the US. Through ownership and participation in shareholder voting, these investors may exert influence on corporate governance. Research has shown that large institutional blockholders exhibit higher propensities to support management in shareholder voting than other shareholders, which may be interpreted as evidence that these investors have a preference towards management and thus attach a private value to management-supported outcomes.

This interpretation, however, may conflate preferences and information. Investors vote without observing a proposal's true effect on firm value (its common value); a rational investor forms beliefs about whether the proposal's passing will increase or decrease firm value by observing public information (via proxy statements and advisor recommendations, for example), their own private information, and information embedded in the voting environment itself. As a result, the same investor may vote differently on otherwise similar proposals than their unconditional preference for the proposal would suggest.

In this paper, we develop a structural model in which firms are held by large blockholders and smaller, dispersed shareholders who vote on management-sponsored proposals. Proposal quality is uncertain at the time of the vote. Shareholders have heterogeneous information about proposal quality and may also have heterogeneous preferences for passage (i.e., private values), modeled as an extra utility from the proposal passing (or failing) that is independent of the proposal's effect on firm value (i.e., its common value).[1] In equilibrium, shareholders follow cutoff strategies: each shareholder supports the proposal when their posterior belief exceeds an endogenous, shareholder-specific threshold. We use the model to study how preferences and information jointly shape equilibrium voting and observed voting outcomes.

A stylized example clarifies the model's intuition and empirical relevance. A blockholder who supports more proposals than smaller shareholders may appear biased toward management. But a strategic voter conditions on the event that her vote is decisive, and this pivotal event is itself informative about proposal quality. If other shareholders are more skeptical of

[1] In the model and estimation, we allow blockholders to have heterogeneous private values and private information quality, while we assume that smaller, dispersed shareholders share the same private value and information quality.

management proposals, a close vote means many received favorable signals to nearly pass the proposal, which is more likely when the proposal is good. Pivotality therefore conveys that the proposal is more likely to be good, and the blockholder rationally supports more proposals without any preference bias.

When signals are correlated (for example, through proxy advisor reports), the strategic effect attenuates. Shareholders draw on the same public information, so a close vote partly reflects that shared signal rather than independent private information, and the pivotal event is less informative. As a result, the same blockholder voting in favor would likely be more indicative of a preference towards management.

The identification problem extends beyond the stylized example: because both public information and strategic conditioning shape equilibrium voting, identical voting data can be consistent with different combinations of preferences and information.[2] Our main goal in this paper is to provide a solution to this identification problem and jointly estimate shareholder's preferences and information.

Identification proceeds as follows. First, we use proxy advisor recommendations as a proxy for public information. Recommendations broadly separate proposals into those public information indicates are good and those it indicates are bad. Second, residual cross-blockholder voting covariances within these groups help separate public from private information, since shared exposure to common public signals is the source of co-movement in blockholder votes.

With the information environment identified, preferences are backed out from each shareholder's equilibrium cutoff condition, which embeds how strategic conditioning on pivotality shapes the cutoff that rationalizes observed voting. As such, the shareholder-specific preferences (private values) we recover are independent of any information effects, which lets us distinguish blockholders who vote for a proposal because it is good from those who simply prefer to support management. We formalize this argument and prove that preferences and the information environment are jointly identified.

[2]We formalize this observation: a continuum of information and preference parameters generate identical individual voting patterns. That is, for any candidate parameter vector, it is possible to find a different parameter vector that generates the same marginal distributions of dispersed and blockholder support. Identification requires both structure on the information environment and moments beyond individual voting patterns. The source of non-identification, like in the example above, is that correlated information and strategic voting jointly determine the mapping from preferences to votes.

This identification strategy connects to the broader empirical literature on shareholder voting. One strand of this literature studies how heterogeneous information across shareholders shapes voting outcomes while taking shareholder private values as given; another studies private values (preferences or biases) while taking the information environment as given. Both have advanced our understanding of how institutional investors vote, and we contribute to this literature by formally establishing, within a structural model of strategic voting, that the information environment and private values are jointly identified from observed voting data, and by providing an estimation strategy that recovers both.

We estimate the model on Say-on-Pay (executive compensation) proposals, with six large mutual fund blockholders as the modeled actors: Vanguard, BlackRock, State Street, Dimensional, Fidelity, and T. Rowe Price. ISS recommendations serve as the observed proxy for public information, and we recover latent preferences and information parameters via the Generalized Method of Moments (GMM).

Our estimation procedure recovers two sets of objects for each shareholder: information parameters governing the relative precision of private and common signals, and private value parameters that are corrected for information effects. Because equilibrium voting depends on beliefs formed from both public signals and inference about other shareholders' behavior, the recovered private values reflect information-adjusted preferences for management proposals rather than raw vote propensities. The resulting estimates imply sizable differences between observed and preference-implied support: while most blockholders exhibit high unconditional approval rates in the data, their preference-implied support is closer in magnitude to that of dispersed shareholders.[3]

These results highlight one of the model's central predictions: higher observed support does not necessarily reflect a stronger preference for passage. Strategic effects and information asymmetry generate an equilibrium in which blockholders with more precise signals or greater voting weight may appear more supportive of management even when their underlying preferences do not imply it.

[3]Dimensional is a notable exception, with systematically lower raw support consistent with its distinct investment philosophy. It is reassuring for the model that the same framework can rationalize and fit this qualitatively different voting pattern alongside the Big Three and other large funds.

A central question in the governance literature is whether blockholders vote differently from other shareholders because of differences in preferences or in information (Hu et al., 2024; Matsusaka and Shu, 2022; Bubb and Catan, 2022; Bolton et al., 2020). We implement counterfactuals that separately equalize information and preferences across investor types by assigning blockholders the same public and private information precision, and separately the same preferences, as dispersed shareholders.

Equalizing information quality has a far larger effect on voting outcomes than equalizing preferences: the close vote rate nearly doubles relative to the baseline. This arises because (more) precise private signals allow blockholders to better distinguish good from bad proposals, pushing outcomes away from the passing threshold. Equal information quality removes this state-contingency and blockholders vote more uniformly on their pro-management baseline, supporting more bad proposals and fewer good ones. Equalizing preferences, by contrast, has a smaller effect because blockholders' precise signals preserve the state-contingency of their votes regardless of the preference level. In this sense, differences in information matter much more than differences in preferences for voting outcomes.

The paper is organized as follows. The rest of this section discusses the literature. Section 2 outlines the general model. Section 3 analyzes the model and illustrates the identification problem induced by public information and strategic voting. Section 4 presents the identification and estimation strategies, and describes the data used in the estimation. Section 5 presents results of the estimation. Section 6 concludes.

Literature review. This paper contributes to debates about how large diversified investors affect corporate governance, and to the study of voting under incomplete information. Legal and governance scholarship offers different views of large passive institutions. Bebchuk and Hirst (2019) argue that low-fee business models and potential conflicts can tilt preferences toward passage. Kahan and Rock (2020) and Fisch et al. (2019) emphasize that scale can strengthen stewardship incentives. The analysis here complements these perspectives by showing that observed support rates need not map into preferences once strategic pivotality and information aggregation are taken into account.

Our paper relates to the literature studing shareholder voting under incomplete information. Maug (1999), including Austen-Smith and Banks (1996), Feddersen and Pesendorfer (1997), and Levit and Malenko (2011). We incorporate blockholder ownership and allow for a preference-for-passage component, and We study how correlated signals attenuate the informativeness of the pivotal event. The interpretation of correlation follows Malenko and Malenko (2019), who analyze the effect of proxy-advisor recommendations on information aggregation. Bar-Isaac and Shapiro (2020) examine participation versus abstention with blockholders; Levit et al. (2025) study trading before the vote with public information; Malenko et al. (2025) consider the design of proxy-advisor products. Our focus is on how preferences and information jointly determine equilibrium cutoff strategies and observed support.

Empirical work has sought to measure investor philosophies from votes. Bubb and Catan (2022) use principal components to identify dimensions of governance preferences; Bolton et al. (2020) place mutual fund families along an ideological spectrum; Yi (2021) uses a Bayesian approach to characterize preferred governance structures. Our approach is complementary. We recover preference-for-passage and information parameters by estimating equilibrium strategies, which directly adjusts for strategic and informational interactions.

2. Model

2.1. Ownership and Voting

Each firm j is owned by a set of investors, each indexed by i. Blockholders $b \in \{1, \ldots, N_B\}$ each own a fraction ψ_b of outstanding shares, such that total blockholder ownership is $\Psi_B = \sum_{b=1}^{N_B} \psi_b$. The remaining shares $\Psi_D = 1 - \Psi_B$ are held by N_D symmetric dispersed shareholders, each of whom owns a fraction $\psi_D = \Psi_D / N_D$.

The firm's management puts up a proposal of unknown quality on which shareholders vote. Examples include advisory votes on executive compensation or director elections. Proposal quality is captured by a common-value state,

$$x_j \sim N(0, 1), \tag{1}$$

such that x_j is unknown to shareholders at the time of voting but has a prior distribution that is common knowledge.[4] Management provides a recommendation for the proposal; voting in favor corresponds to supporting management's recommendation.

Each shareholder is characterized by a preference parameter $\delta_i \in \mathbb{R}$. Let $\Delta \equiv \{\delta_i\}$ denote the (common knowledge) set of preferences for all shareholders. Shareholders with positive δ_i receive greater utility when the proposal passes; that is, they prefer passage of proposals regardless of their intrinsic quality. We interpret investors with positive δ_i as being more inclined to support management. However, a positive δ_i need not represent any literal bias towards management, it simply captures any preference that makes adoption more attractive, independent of its quality.

Given the outcome of the vote $\mathrm{P}_j \in \{0, 1\}$, shareholder i's payoff is

$$U_i(\mathrm{P}_j, x_j, \delta_i) = \mathrm{P}_j \times (x_j + \delta_i), \tag{2}$$

in which payoffs are normalized to zero when the proposal fails. When the proposal passes, the payoff reflects two components: the common value x_j, which represents the proposal's impact on firm value and the private value δ_i. All dispersed shareholders share the same preference parameter δ_D, while each blockholder $b \in \{1, \ldots, N_B\}$ has its own δ_b. The complete preference environment is thus $\Delta = \{\delta_D, \{\delta_b\}_{b=1}^{N_B}\}$. Note that

$$\Pr(x_j + \delta_i > 0) = 1 - \Phi(-\delta_i) = \Phi(\delta_i), \tag{3}$$

where $\Phi(\cdot)$ is the standard normal CDF. This quantity gives each shareholder's baseline approval probability, absent any effect of (private or public) information. We refer to $\Phi(\delta_i)$ as shareholder i's preference-implied support rate.

We assume full vote turnout for analytical clarity, though when we estimate the model we allow for abstentions. The proposal passes, $\mathrm{P}_j = 1$, if the total support rate exceeds the passing threshold λ^*, and fails, $\mathrm{P}_j = 0$, otherwise. Under a majority rule, for example, $\lambda^* = 0.5$,

[4] The standard normal is a normalizing assumption. Since shareholders observe only signals about x_j, the mean and variance of the prior are not identified separately from signal precision, which we introduce below; fixing $x_j \sim N(0, 1)$ is without loss of generality conditional on normality. Across proposal types, the distribution of proposal quality may differ, so we focus on a particular proposal type in estimation.

whereas under a supermajority $\lambda^* > 0.5$.[5]

Let $V_{bj} \in \{0,1\}$ denote blockholder b's vote, and define the vector of blockholder votes $\mathbf{V}_{Bj} = [V_{1j}, \ldots, V_{N_B j}]$ and the vector of ownership weights $\mathbf{\Psi}_B = [\psi_1, \ldots, \psi_{N_B}]$. Given $\mathbf{V}_{Bj}$, the share of total votes cast in favor by blockholders is $s_{Bj} = \mathbf{V}_{Bj}\mathbf{\Psi}'_B$. Let τ_j denote the fraction of dispersed shareholders who vote for the proposal. The total support rate is therefore

$$\lambda_j = s_{Bj} + \tau_j = \mathbf{V}_{Bj}\mathbf{\Psi}'_B + \tau_j \Psi_D, \tag{4}$$

and the proposal passes if $\lambda_j \geq \lambda^*$.

2.2. Information Environment

Before voting, shareholders receive noisy signals about the proposal which causes them to update their beliefs about underlying quality x_j.

2.2.1. Public signals

We assume that before voting a proxy advisor produces a report about the proposal. We model this report as a continuous, noisy signal of the underlying state:

$$s_j = x_j + u_{Ij}, \qquad u_{Ij} \sim N(0, \sigma_I^2), \tag{5}$$

where σ_I^2 represents the quality of the proxy advisor's information. We interpret s_j as the proxy advisor's underlying research report that shareholders may access, but we assume that dispersed and block shareholders see different versions of the report. More concretely, these two types of shareholder form group-specific noisy public assessments of proposal quality based on this report and other public information. Dispersed shareholders use

$$\eta_{Dj} = s_j + \upsilon_{Dj}, \qquad \upsilon_{Dj} \sim N(0, \sigma_{\upsilon D}^2), \tag{6}$$

[5] We abstract from strategic turnout (Zachariadis et al., 2020). Additionally, under the normal structure, the passing threshold λ^* is absorbed into the preference parameters δ_i, so simple majority and supermajority rules are equivalent up to reparameterization.

while blockholders use

$$\eta_{Bj} = s_j + v_{Bj}, \qquad v_{Bj} \sim N(0, \sigma^2_{vB}). \tag{7}$$

The group-specific noise terms v_{Dj} and v_{Bj} capture differences in how precisely dispersed shareholders and blockholders can observe or process the proxy advisor's report and related public disclosures. Since $s_j = x_j + u_{Ij}$, we can equivalently write the public signals in reduced form as

$$\eta_{Dj} = x_j + u_{Dj}, \qquad u_{Dj} := u_{Ij} + v_{Dj},$$
$$\eta_{Bj} = x_j + u_{Bj}, \qquad u_{Bj} := u_{Ij} + v_{Bj},$$

with u_{Ij}, v_{Dj}, and v_{Bj} mutually independent and independent of x_j. Hence

$$\sigma^2_{u_D} = \sigma^2_I + \sigma^2_{vD}, \qquad \sigma^2_{u_B} = \sigma^2_I + \sigma^2_{vB},$$

and the precisions $1/\sigma^2_{u_D}$ and $1/\sigma^2_{u_B}$ summarize the effective quality of public information for dispersed and block shareholders, respectively.

Imperfectly correlated public information across block and dispersed shareholders. Allowing η_{Dj} and η_{Bj} to be imperfectly correlated, rather than imposing a single common public signal, is both empirically and conceptually important. Empirically, we observe residual cross-blockholder voting covariance even after conditioning on dispersed support and the proxy advisor recommendation, which would not arise if blockholders' public information were fully captured by what dispersed shareholders observe. Conceptually, with a single common public signal, dispersed support (in the limit as dispersed shareholders become atomistic) would itself become a sufficient statistic for the public information and would absorb any independent role for η_{Bj} in blockholder votes. Allowing σ^2_{vD} and σ^2_{vB} to differ accommodates the natural idea that blockholders and dispersed shareholders process the same underlying report with different precision.

Binary proxy advisor recommendations. It is useful to note that, empirically, we observe a binary recommendation. We assume that this binary recommendation arises from the continuous report such that

$$S_j = \mathbb{1}\left[s_j > \xi\right],$$

where ξ is a fixed threshold which can be interpreted as the headline "For/Against" recommendation of a proxy advisor such as ISS.[6] In the data, the econometrician observes this binary signal S_j ex post, but not the underlying continuous report s_j or the separate noise components in investors' public signals. We assume that shareholders do not directly observe S_j; instead, we summarize the effect of the proxy advisor's analysis through the reduced-form public signals η_{Dj} and η_{Bj} and use the observed binary recommendation S_j in the estimation to help identify the information environment.[7]

2.2.2. Private signals

In addition to the public signals, each shareholder i receives a private signal

$$z_{ij} = x_j + \varepsilon_{ij},$$

where $\varepsilon_{ij} \sim N(0, \sigma^2_{\varepsilon_i})$ is independent across shareholders and proposals and independent of x_j and all public signal shocks. This private signal represents investor-specific information or analysis that may not be reflected in the public signals. Differences in the precision of these private signals, $1/\sigma^2_{\varepsilon_i}$, capture heterogeneity in shareholders' information quality.

[6] We remain agnostic about whether the proxy advisor's recommendation rule is biased toward or against management. Because ξ is a fixed threshold, any constant bias is absorbed into its estimated value, and our identification strategy goes through as long as the bias does not vary across proposals.

[7] Incorporating S_j into shareholders' information sets would replace Gaussian updating with a truncated-normal posterior, since η_{Dj} does not render S_j redundant. Our assumption that shareholders rely on their own assessment of public information rather than the headline recommendation avoids this complication. Even if shareholders are aware of the binary recommendation, its marginal information content conditional on their detailed assessment of public disclosures is likely small. Additionally, because the econometrician observes only the binary recommendation S_j and voting outcomes, the data identify only the composite variances $\sigma^2_{u_D}$ and $\sigma^2_{u_B}$; we cannot separately recover σ^2_I and the group-specific components σ^2_{vD} and σ^2_{vB}. We return to this point in the estimation section.

Dispersed shareholders share a common private signal precision,

$$\sigma^2_{\varepsilon_i} = \sigma^2_{\varepsilon_D} \quad \text{for all dispersed } i, \tag{8}$$

reflecting that small investors rely on similar, relatively noisy sources of information. By contrast, each blockholder b may have a distinct information precision,

$$\sigma^2_{\varepsilon_i} = \sigma^2_{\varepsilon_b} \quad \text{for blockholder } b. \tag{9}$$

This structure implies that all shareholders start from a common prior and update using both public and investor-specific information. Dispersed shareholders observe (η_{Dj}, z_{ij}) and blockholders observe (η_{Bj}, z_{bj}). The shared component s_j induces correlation in posterior beliefs across shareholders.

2.3. Voting Strategy

Given the information environment described above, each shareholder observes a group-specific continuous public signal $\eta_{ij} \in \{\eta_{Dj}, \eta_{Bj}\}$, as well as her private signal z_{ij}. Conditional on these signals, shareholder i forms a posterior belief about the latent state x_j using standard Gaussian updating. Shareholder i votes in favor of the proposal if and only if her posterior expectation of proposal quality exceeds a cutoff k_i:

$$V_{ij} = 1 \quad \Longleftrightarrow \quad \mathbb{E}[x_j \mid \eta_{ij}, z_{ij}] > k_i. \tag{10}$$

We can think of k_i as the shareholder's decision threshold: shareholders with lower k_i are more likely to support proposals, whereas those with higher k_i require higher proposal quality to vote in favor. Importantly, k_i and δ_i need not move in unison, as we show below.

Because both the continuous public signal η_{ij} and the private signal z_{ij} are normally distributed around x_j, the posterior distribution of x_j given (η_{ij}, z_{ij}) is normal with mean m_{ij} and variance $\sigma^2_{x|\eta,z}$. Specifically,

$$x_j \mid (\eta_{ij}, z_{ij}) \sim N\left(m_{ij},\ \sigma^2_{x|\eta,z}\right),$$

where the posterior mean for shareholder i is

$$m_{ij} \equiv \mathbb{E}[x_j \mid \eta_{ij}, z_{ij}] = \frac{\dfrac{\eta_{ij}}{\sigma_{u_i}^2} + \dfrac{z_{ij}}{\sigma_{\varepsilon_i}^2}}{\Lambda_i}, \qquad \Lambda_i \equiv 1 + \frac{1}{\sigma_{u_i}^2} + \frac{1}{\sigma_{\varepsilon_i}^2} \tag{11}$$

Here $\sigma_{u_i}^2$ is the variance of the reduced-form public signal noise relevant for shareholder i, and $\sigma_{\varepsilon_i}^2$ is the variance of her private signal noise. Conditional on the true proposal quality x_j and the public signal η_{ij}, the only source of randomness is the idiosyncratic noise ε_{ij} in the private signal. Substituting $z_{ij} = x_j + \varepsilon_{ij}$ and noting that $\varepsilon_{ij} \sim N(0, \sigma_{\varepsilon_i}^2)$ gives:

$$m_{ij} \mid (\eta_{ij}, x_j) \sim N\left(\frac{\dfrac{\eta_{ij}}{\sigma_{u_i}^2} + \dfrac{x_j}{\sigma_{\varepsilon_i}^2}}{\Lambda_i}, \ \frac{1}{\sigma_{\varepsilon_i}^2 \Lambda_i^2}\right).$$

The probability that shareholder i supports the proposal, conditional on (x_j, η_{ij}), is

$$\Pr(V_{ij} = 1 \mid x_j, \eta_{ij}) = \Pr\left(m_{ij} > k_i \mid x_j, \eta_{ij}\right) = \Phi\left(\sigma_{\varepsilon_i}\left[\frac{\eta_{ij}}{\sigma_{u_i}^2} + \frac{x_j}{\sigma_{\varepsilon_i}^2} - k_i\Lambda_i\right]\right), \tag{12}$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. Equation (12) shows that a shareholder is more likely to vote in favor when her public signal η_{ij} is high, when proposal quality x_j is high, or when her cutoff k_i is low. The terms inside the brackets reflect the information-based component of the voting rule and each signal is weighted by its precision. The final term, $k_i\Lambda_i$, captures differences in voting thresholds and thus differences in baseline support rates. Heterogeneity in public signal precisions ($\sigma_{u_i}^2$), private signal precisions ($\sigma_{\varepsilon_i}^2$), and thresholds k_i generates cross-sectional variation in both the level and the responsiveness of voting behavior. This variation is central for identification in the empirical estimation.

2.4. Equilibrium

In this section, we lay out the components that determine the equilibrium and then define the equilibrium.

2.4.1. Equilibrium components

Pivotality. Each shareholder votes strategically, recognizing that her action affects payoffs only in states where it changes the voting outcome. In other words, although a shareholder does not know ex ante whether she will be pivotal, she chooses her vote as if at the pivotal margin and focuses on states in which the overall outcome is close (Bond and Eraslan, 2010).

Let the total support rate be defined by the function

$$\lambda(v_i, V_{Bj}^{-i}, \tau_j) = \psi_i v_i + \Psi_{\text{for}}(V_{Bj}^{-i}) + \Psi_D \tau_j,$$

where $v_i \in \{0, 1\}$ is shareholder i's vote, ψ_i is her voting weight, V_{Bj}^{-i} collects blockholders' votes (excluding i if i is a blockholder), $\Psi_{\text{for}}(V_{Bj}^{-i}) \equiv \sum_{\ell \neq i} \psi_\ell V_{\ell j}$, and τ_j is the fraction of dispersed shareholders supporting the proposal. The proposal passes if and only if $\lambda_j(\cdot) \geq \lambda^*$.

Shareholder i is pivotal if her vote flips the outcome:

$$\text{piv}_{ij} \;\equiv\; \{\lambda(1, V_{Bj}^{-i}, \tau_j) \geq \lambda^*\} \;\cap\; \{\lambda(0, V_{Bj}^{-i}, \tau_j) < \lambda^*\}.$$

For a blockholder b with weight $\psi_b > 0$, conditional on how other blockholders vote, pivotality occurs when dispersed support makes it such that b's vote would flip the outcome:

$$\text{piv}_{iv} = \tau_j \in \left[\frac{\lambda^* - \psi_b - \Psi_{\text{for}}(V_{Bj}^{-b})}{\Psi_D}, \; \frac{\lambda^* - \Psi_{\text{for}}(V_{Bj}^{-b})}{\Psi_D}\right),$$

where $\Psi_{\text{for}}(V_{Bj}^{-b})$ is the aggregate support rate of all blockholders excluding b. For a dispersed shareholder i with ownership weight ψ_D, pivotality means that, holding fixed all other votes, i's vote flips the outcome. Let $\tau_{-i,j}$ denote the fraction of dispersed shareholders other than i who vote in favor. Then i is pivotal iff

$$\begin{aligned}
&\Psi_{\text{for}}(V_{Bj}) + \Psi_D\,\tau_{-i,j} \;<\; \lambda^* \;\leq\; \Psi_{\text{for}}(V_{Bj}) + \Psi_D\,\tau_{-i,j} + \psi_D \qquad \Longleftrightarrow \\
&\tau_{-i,j} \in \left[\frac{\lambda^* - \Psi_{\text{for}}(V_{Bj}) - \psi_D}{\Psi_D}, \; \frac{\lambda^* - \Psi_{\text{for}}(V_{Bj})}{\Psi_D}\right).
\end{aligned}$$

Pivotality is investor-specific and depends on the realized voting environment. Hence, the conditioning event relevant for an investor's best response is itself endogenous to her voting weight, along with her beliefs about the information and preferences of other shareholders. Voting decisions are therefore interdependent even conditional on observables: each investor's optimal action depends on her own information and on her beliefs about how others map information and preferences into votes.

Best responses. When deciding how to vote, each shareholder compares the expected payoff from supporting the proposal to that from opposing it, recognizing that her vote only affects the outcome in states where she is pivotal. Conditional on being pivotal and observing a posterior mean exactly equal to her cutoff, $m_{ij} = k_i$, each shareholder is indifferent between voting "For" and "Against," which is formalized in the following best response condition:

$$\mathrm{E}\left[\, x_j \;\middle|\; \eta_{ij},\ m_{ij} = k_i,\ \mathrm{piv}_{ij} \right] = -\,\delta_i, \tag{13}$$

where δ_i is her innate preference towards the proposal passing. Equivalently, expressing the indifference condition in terms of the private signal, we may write

$$\mathrm{E}\left[\, x_j \;\middle|\; \eta_{ij},\ z_{ij} = z^\star_{ij}(\eta_{ij}),\ \mathrm{piv}_{ij} \right] = -\,\delta_i, \tag{14}$$

where the cutoff private signal realization $z^\star_{ij}(\eta_{ij})$ solves $m_{ij} = k_i$ and is given by

$$z^\star_{ij}(\eta_{ij}) = \sigma^2_{\varepsilon_i}\left(\Lambda_i\, k_i - \frac{\eta_{ij}}{\sigma^2_{u_i}}\right), \qquad \Lambda_i = 1 + \frac{1}{\sigma^2_{u_i}} + \frac{1}{\sigma^2_{\varepsilon_i}}.$$

At the pivotal margin, the shareholder is willing to vote "For" only if the expected quality of the proposal, conditional on observing exactly the cutoff signal and on being pivotal, offsets her preference δ_i. Equation (14) therefore characterizes the equilibrium cutoff k_i that makes shareholder i just indifferent when her vote is decisive.

Signal and strategic effects. Expanding the expected payoff conditional on being pivotal yields

$$\mathrm{E}[x_j \mid \eta_{ij}, z_{ij}, \mathrm{piv}_{ij}] = \int_x x_j\, f\left(x_j \mid \eta_{ij}, z_{ij}, \mathrm{piv}_{ij}\right)\, dx,$$

where the posterior density of proposal quality can be decomposed into a signal component and a strategic component:

$$f\left(x_j \mid \eta_{ij}, z_{ij}, \mathrm{piv}_{ij}\right) \propto f(x_j) \underbrace{f(\eta_{ij} \mid x_j)}_{\text{public signal effect}} \underbrace{f(z_{ij} \mid x_j)}_{\text{private signal effect}} \underbrace{\Pr\left(\mathrm{piv}_{ij} \mid x_j, \eta_{ij}\right)}_{\text{strategic effect}}. \tag{15}$$

The first two braced terms form the signal effect: they summarize what the shareholder learns about proposal quality from her own signals, (η_{ij}, z_{ij}). Because dispersed shareholders observe η_{Dj} and all blockholders observe the common blockholder signal η_{Bj}, the public signal component is group-specific.

The final term, $\Pr(\mathrm{piv}_{ij} \mid x_j, \eta_{ij})$, is the strategic effect. It captures the additional information contained in the event of being pivotal. This probability depends on how other shareholders vote, and therefore on their equilibrium strategies, their private information $(\sigma_{-i}, \delta_{-i})$, and on the public signal they observe. Since shareholder i does not observe the other group's public signal (e.g., a dispersed shareholder does not observe η_{Bj}), the strategic effect implicitly integrates over the unobserved public signals of the other investor groups. Conditioning on pivotality therefore shifts beliefs toward states of the world in which the aggregate vote is close, providing information about x_j beyond what is contained in (η_{ij}, z_{ij}).

2.4.2. Posterior beliefs about proposal quality

Determining equilibrium voting strategies requires evaluating the posterior in (15) for each shareholder. This, in turn, requires computing each shareholder's probability of being pivotal for each possible value of proposal quality x_j and her observed public signal: η_{Dj} for dispersed shareholders and η_{Bj} for blockholders. Following Myatt (2015) and Zachariadis et al. (2020), we derive the limit of the game as N_D grows large. The formal expressions are given in Lemma A.1 and Lemma A.2 in the appendix; here we describe the key intuition for each investor type.

Dispersed shareholders. In the large-N_D limit, the dispersed support rate is a deterministic function of (x_j, η_{Dj}). A dispersed shareholder is therefore pivotal only at knife-edge values of x_j for which the implied support rate exactly equals the threshold needed for passage, given each possible blockholder vote profile V_{Bj}. The posterior in (15) concentrates on these weights, each weighted by the conditional prior $f(x_j \mid \eta_{Dj})$, the private signal likelihood $f(z_{Dj} \mid x_j)$, and the probability of the blockholder vote profile that makes the state pivotal, integrated over the unobserved blockholder public signal η_{Bj}.

Blockholders. For a blockholder b with voting weight ψ_b, pivotality occurs when dispersed support falls in an interval $[\tau_L^{V_{B,-b,j}}, \tau_H^{V_{B,-b,j}}]$, not at a single point. Pivotality therefore has positive probability conditional on x_j. The blockholder's posterior in (15) combines the prior shaped by the blockholder public signal $f(x_j \mid \eta_{Bj})$, the private signal likelihood $f(z_{bj} \mid x_j)$, the pivotality probability $\Pi^{V_{B,-b,j}}(x_j, \eta_{Bj})$ integrated over the unobserved dispersed public signal η_{Dj} conditional on both x_j and η_{Bj}, and the strategic likelihood of the other blockholders' vote profile $\Pr(V_{B,-b,j} \mid x_j, \eta_{Bj})$. The conditioning on η_{Bj} in the pivotality probability reflects the latent correlation between the dispersed and blockholder public signals induced by the common proxy advisor report. Conditioning on pivotality shifts the blockholder's beliefs toward states in which dispersed support is close to the passage threshold, providing information about x_j beyond what is contained in (η_{Bj}, z_{bj}). The equilibrium cutoff k_b is then pinned down by the indifference condition (13) evaluated under this posterior.

2.4.3. Equilibrium Definition

Maintaining the information and preference environments from above, the equilibrium concept is pure-strategy Bayesian–Nash: each shareholder chooses a cutoff strategy that maximizes expected utility given her information set (which conditions on the state of the world when she is pivotal), along with her innate preference towards the proposal passing. We define the equilibrium as the limit of the symmetric equilibrium of the finite game as $N_D \to \infty$; in particular, we treat the dispersed shareholder base as atomistic and derive the limiting equilibrium conditions in Appendix A.

Each dispersed shareholder observes (η_{Dj}, z_{dj}), while blockholder b observes (η_{Bj}, z_{bj}).

Each shareholder votes "For" if the expected quality of the proposal at the pivotal margin, net of her private preference, is nonnegative. In equilibrium this can be expressed as a cutoff rule in terms of her posterior mean m_{ij}, for (11):

$$\mathrm{E}\left[x_j \mid \eta_{Dj}, z_{Dj}, \mathrm{piv}_{Dj}\right] + \delta_D \geq 0 \iff m_{Dj} \geq k_D, \tag{16}$$

$$\mathrm{E}\left[x_j \mid \eta_{Bj}, z_{bj}, \mathrm{piv}_{bj}\right] + \delta_b \geq 0 \iff m_{bj} \geq k_b, \quad \forall b \in \{1, \dots, N_B\}, \tag{17}$$

where k_D is the common dispersed cutoff and k_b is blockholder b's cutoff. The effect of pivotality is captured entirely by these equilibrium cutoffs. The equilibrium can be equivalently written as a system of indifference conditions at the cutoffs:

$$\mathrm{E}\left[x_j \mid \eta_{Dj},\ m_{Dj} = k_D,\ \mathrm{piv}_{Dj}\right] = -\,\delta_D,$$

$$\mathrm{E}\left[x_j \mid \eta_{Bj},\ m_{bj} = k_b,\ \mathrm{piv}_{bj}\right] = -\,\delta_b, \quad \forall b \in \{1, \dots, N_B\}.$$

Cutoffs k_i therefore reflect shareholders' strategic inference under the event of being pivotal, which depends on the aggregate support implied by others' strategies and information.

Equilibrium Support Rates. Given an equilibrium cutoff profile $\mathbf{k} = \left\{k_D, \{k_b\}_{b=1}^{B}\right\}$, the (prior) probability that shareholder i votes "For", i.e., her equilibrium support rate, is

$$\Pr(V_{ij} = 1) = \Pr\left(m_{ij} \geq k_i\right) = \iint 1\left\{m_{ij}(\eta, z) \geq k_i\right\} f_i(\eta)\, f_i(z \mid \eta)\, d\eta\, dz, \tag{18}$$

where $f_i(\eta)$ and $f_i(z \mid \eta)$ denote the public and private signal densities implied by $(\sigma^2_{u_i}, \sigma^2_{\varepsilon_i})$ in the information environment.[8] Thus, equilibrium support rates are determined by the distribution of each shareholder's posterior mean relative to her cutoff, with the cutoffs themselves pinned down by the pivotality-adjusted indifference conditions.

[8]For dispersed shareholders, $(\sigma^2_{u_i}, \sigma^2_{\varepsilon_i}) = (\sigma^2_{u_D}, \sigma^2_{\varepsilon_D})$ and $\eta = \eta_{Dj}$; for blockholder b, $(\sigma^2_{u_i}, \sigma^2_{\varepsilon_i}) = (\sigma^2_{u_B}, \sigma^2_{\varepsilon_b})$ and $\eta = \eta_{Bj}$.

3. Model Analysis

In this section, we undertake comparative statics of key parameters, to understand the model's forces. In particular, we study how preferences, information, and strategic behavior shape voting outcomes and how these forces affect the interpretation of observed support rates. We focus on a special case where there is one blockholder holding 20% of the ownership, and the rest of the ownership is held by dispersed shareholders.

3.1. Decomposing the Observed Blockholder Support rate

We conduct a comparative static exercise to isolate and illustrate how the blockholder's information precision and strategic voting considerations impact her observed voting behavior, what we refer to as the *signal* and *strategic* effects in the model.[9] This allows us to study how changes in observed voting behavior can reflect improvements in the blockholder's information quality or shifts in her strategic incentives from conditioning on being pivotal, two channels that may be confounded empirically.

To separate these effects, consider two objects. The first is the blockholder's equilibrium support rate,

$$\tau_b^S = \Pr(m_b \geq k_b)\,,$$

where m_b is the blockholder's posterior mean (as in Eq. 11), and k_b is her equilibrium voting threshold. This embeds both the signal and strategic effect. The second is the non-strategic support rate,

$$\tau_b^{NS} = \Pr(m_b \geq -\delta_b)\,,$$

which is the support rate of the blockholder if she did not condition on pivotality. The difference in these two objects as we vary parameters isolates the impact of strategic voting on the blockholder's equilibrium decision. As comparison, we also consider the blockholder's

[9] The signal effect is simply how the quality of information received (holding strategic effects constant) influences a shareholder's posterior beliefs about proposal quality. The strategic effect arises from how conditioning on being pivotal alters these beliefs in equilibrium. See equation 15.

preference-implied support rate,

$$\Phi(\delta_b) = \Pr\left(x_j \geq -\delta_b\right),$$

which is the support rate if the blockholder voted purely by her private value (see Eq. 3).

In the analysis, under the assumed parameters , we vary the precision of the blockholder's private information under three different values of their private value: $\delta_b \in \{0, 0.5, -0.5\}$. We solve for the equilibrium cutoffs, given the best response of dispersed shareholders and compare the three support rates described above. In this exercise, we assume noisy public signals for both shareholders to focus on the effect of private information.[10]

Figure 1 illustrates these forces for three values of δ_b. When $\delta_b = 0$, the non-strategic support rate is constant at 50%, since the blockholder supports whenever her posterior exceeds her prior. In this case, the strategic support rate lies above 50% when private information is imprecise and converges back to 50% as private information becomes more precise. When $\delta_b = 0.5$, the non-strategic support rate lies above the preference-implied benchmark $\Phi(\delta_b)$, while the strategic support rate lies below it, and the ordering is reversed for $\delta_b = -0.5$.

Across all three panels, the main pattern is that, for $\delta_b \neq 0$, the non-strategic and strategic support rates lie on opposite sides of the preference-implied benchmark $\Phi(\delta_b)$, and both converge toward that benchmark as private information becomes more precise. Under non-strategic voting, imprecise private information magnifies the effect of bias, pushing support away from $\Phi(\delta_b)$.[11]

Strategic voting offsets bias because, when signals are imprecise, the pivotality event is informative about dispersed shareholders' signals, and this informativeness dissipates as the blockholder's information quality improves. For the blockholder to be pivotal, dispersed shareholders must, on average, have received less favorable signals when she is biased toward passage ($\delta_b > 0$), and more favorable signals when she is biased against passage ($\delta_b < 0$).

[10] We restrict attention to noisy public signals so that variation in private-signal precision drives the comparative statics. With precise public signals, equilibrium cutoffs respond to both signal types, and the comparative static in private-signal precision can be non-monotone.

[11] Given the model, we can write $\tau_b^{NS} = \Phi\left(\delta_b \sqrt{\frac{\pi_b}{\pi_b - 1}}\right)$. Since $\frac{\pi_b}{\pi_b - 1} > 1$ for finite π_b, imperfect private information pushes the support rate away from $\Phi(\delta_b)$, with convergence to $\Phi(\delta_b)$ as $\pi_b \to \infty$.

As such, observed blockholder voting reflects both preference and information quality, with strategic effects operating through the informational content of pivotality, holding fixed the voting behavior and parameters of the other shareholders.

3.2. Information and Observational Equivalence

Given that the support rates are determined by preferences, information environment, and strategic behavior, this subsection illustrates a central implication of the model: the same observed support rate can be rationalized by different underlying information and preference environments.

In this comparative static exercise, we hold fixed the blockholder's and dispersed shareholders' support rates, and vary the precision of the blockholder's private signal, recovering the preference parameter needed to sustain the same voting outcome in equilibrium. This mirrors the exercise above, but with the support rate rather than the preference parameter held fixed. The preceding comparative static then shows how the recovered preference must adjust as private information changes. To implement the analysis, we set the blockholder's observed support rate to be 75%, vary the blockholder's private signal precision, and back out the δ_b consistent with that support rate. We hold dispersed parameters fixed.

Figure 2 presents the results. The horizontal line marks the observed support rate held fixed throughout the exercise. Along the horizontal axis, we vary the share of the blockholder's posterior weight placed on private information; along the vertical axis, we report the corresponding preference-implied support rate, $\Phi(\delta_b)$. The dashed curve shows the non-strategic benchmark, while the solid curve shows the strategic benchmark, in which the cutoff is pinned down by the pivotality-adjusted indifference condition.[12] The difference between the curves therefore measures the role of strategic voting in observed support.

Two features of the figure are immediate. First, the non-strategic curve is monotone: as the blockholder's private signal becomes more precise, a higher preference parameter is required to sustain the same observed support rate. This is mechanical effect. With more precise private information, voting depends less on noise and more directly on proposal quality, so

[12]The non-strategic δ_b is the preference parameter that would rationalize the observed support rate if the blockholder ignored pivotality.

the required preference tends toward the observed support rate.

Second, the strategic curve is non-monotone and U-shaped. This reflects the interaction between the same mechanical effect and the informational content of pivotality. In the strategic model, preferences are recovered from $\delta_b = -\mathrm{E}\left[x \mid m_b = k_b,\ \mathrm{piv}_b\right]$, so the recovered preference depends on how the pivotal event shifts belief about proposal quality at the cutoff. When the blockholder's private signal is very imprecise, the recovered strategic preference lies close to the observed support rate. The reason is that the blockholder then relies mainly on public information and on the inference from being pivotal. Holding fixed the dispersed shareholders' voting rule, pivotality is bad news in this region: if the common information were sufficiently favorable, the proposal would tend to pass without the blockholder. Conditioning on pivotality therefore pushes the cutoff upward, so a stronger pro-passage preference is required to sustain the same observed support rate. At intermediate levels of precision, this effect can reverse: being pivotal may instead indicate relatively strong dispersed support at the margin. In that region, pivotality becomes good news and lowers the preference required to sustain the same observed support rate. As private precision rises further, however, the blockholder's own signal increasingly dominates this inference, and the required preference is close to the observed support.

Taken together, the figure shows that the same observed support rate can be rationalized by very different economics. Moreover, because the wedge between the strategic and non-strategic curves can change sign, strategic voting can make investors appear either more or less biased depending on the information environment. This points to a subtle identification problem in separating preference and information environments, which we address in the next section.

4. Identification and Estimation

We now describe the identification and estimation routines of the model from Section 2.[13] The data used for estimation are described in detail in Section 4.4. We suppose that we ob-

[13]Note, we move on from the simplifying assumption in Section 3 that there is a single common information precision and allow dispersed and block shareholders to again have separate common information precisions via σ^2_{uD} and σ^2_{uB}

serve a dataset of proposals indexed by j, containing proposal outcomes and shareholder votes. Specifically, the data include proxy advisor recommendations S_j, realized dispersed shareholder support τ_j, and blockholder-level information on ownership stakes and voting decisions. As we will show, this is sufficient to fully identify the model.

4.1. Identification

For each proposal j, the econometrician observes the proxy advisor's binary recommendation $S_j = \mathbb{1}\left[s_j > \xi\right]$, the dispersed support rate τ_j, and blockholder votes and ownership shares $\{(V_{bj}, \psi_b)\}_{b=1}^{N_B}$ (and hence aggregate blockholder support s_{Bj}). By contrast, proposal quality x_j, the continuous proxy assessment s_j, and the group-specific public signals (η_{Dj}, η_{Bj}) are unobserved. We use the joint cross-sectional distribution of $(S_j, \tau_j, \{V_{bj}\}, \{\psi_b\})$ to identify the information environment Σ and preferences Δ.

Identification rests on three assumptions. First, the cutoff profile $\mathbf{k} = \{k_D, \{k_b\}_{b=1}^{N_B}\}$ arises from a pure-strategy Bayesian–Nash equilibrium and satisfies the model's best-response conditions. Second, the proxy advisor's binary recommendation is monotonic in their underlying report, $S_j = \mathbb{1}\left[s_j > \xi\right]$. Third, the remaining within-S_j variation in dispersed and blockholder voting is fully captured by the latent quality x_j and the public and private signal shocks in the information environment; that is, we assume the information available to shareholders follows the structure outlined in Section 2.

4.1.1. Dispersed and Proxy Advisor Parameters

The joint distribution of (τ_j, S_j) identifies the dispersed information parameters $(\sigma^2_{uD}, \sigma^2_{\varepsilon D})$, and cutoff k_D, along with the correlation $\rho \equiv \mathrm{Corr}(T_j, s_j)$ between the dispersed voting index $T_j = \Phi^{-1}\left(\tau_j\right)$ and the proxy advisor's latent assessment, the proxy advisor threshold ξ. Under the model's assumption of atomistic dispersed shareholders, the dispersed voting rule implies

$$T_j = \sigma_{\varepsilon_D}\left[\frac{\eta_{Dj}}{\sigma^2_{u_D}} + \frac{x_j}{\sigma^2_{\varepsilon_D}} - k_D\Lambda_D\right],$$

so T_j is normally distributed with mean $\mu_T = -\sigma_{\varepsilon_D} k_D \Lambda_D$ and variance σ^2_T that depends on $(\sigma^2_{uD}, \sigma^2_{\varepsilon D})$. The proxy advisor issues a binary recommendation $S_j = \mathbb{1}\left[s_j > \xi\right]$ based on a latent assessment s_j that is jointly normal with T_j. The joint distribution of (T_j, S_j) is fully determined by $(\rho, \xi, \sigma^2_{uD}, \sigma^2_{\varepsilon D}, k_D)$.

Identification proceeds in two steps. First, the probit of S_j on T_j,

$$\Pr(S_j = 1 \mid T_j = t) = \Phi(\gamma_0 + \gamma_1 t),$$

identifies ρ and ξ via $\rho = \gamma_1 / \sqrt{1 + \gamma_1^2}$ and $\xi = -\gamma_0 \sqrt{1 - \rho^2}$. Because S_j is binary, the probit identifies ρ and ξ only up to a common scale normalization on s_j, which we fix by normalizing $\mathrm{Var}(s_j) = 1$.

Second, conditional on (ρ, ξ), the remaining dispersed-side parameters $(\sigma^2_{uD}, \sigma^2_{\varepsilon D}, k_D)$ are identified from the conditional moments of T_j given S_j. The conditional means $\mathrm{E}[T_j \mid S_j = 1]$ and $\mathrm{E}[T_j \mid S_j = 0]$ pin down μ_T and σ_T (since the shift across ISS groups is governed by the known ρ and ξ through the truncated normal). The conditional variances $\mathrm{Var}(T_j \mid S_j = 1)$ and $\mathrm{Var}(T_j \mid S_j = 0)$ then separate public from private dispersed noise: the ISS recommendation truncates only the component of T_j that co-moves with s_j, leaving the independent component unaffected. The relative magnitude of the variance difference across ISS groups therefore identifies the decomposition of σ^2_T into σ^2_{uD} and $\sigma^2_{\varepsilon D}$.

Intuitively, this strategy is analogous to two-stage least squares. By assumption, S_j only impacts dispersed support through the latent proxy advisor report s_j (it is orthogonal to the private signal). The Probit of S_j on T_j is the first stage and identifies the strength and direction of the instrument: ρ. The conditional moments are the second stage as the variation in T_j unexplained by S_j measure the private information channel. Importantly, identifying these parameters allows us to pin down the (econometrician's) posterior $x_j \mid (\tau_j, S_j)$ which converts observables into a sufficient statistic for proposal quality. The remaining variation in blockholder votes then identifies their information environment.

4.1.2. Blockholder Parameters

Conditional on the dispersed-side parameters, we must identify the common blockholder public variance σ^2_{uB} and each blockholder's private variance and cutoff $(\sigma^2_{\varepsilon_b}, k_b)$. Integrating out unobserved signals conditional on (τ_j, S_j) yields, for each blockholder b,

$$\Pr(V_{bj} = 1 \mid \tau_j, S_j) = \Phi\big(\alpha_{b0} + \alpha_{b1} T_j + \alpha_{b2} S_j\big)\,,$$

where $(\alpha_{b0}, \alpha_{b1}, \alpha_{b2})$ are smooth functions of $(\sigma^2_{uB}, \sigma^2_{\varepsilon_b}, k_b)$ and dispersed-side parameters (which we show in the appendix, Lemma B.4).

Since σ^2_{uB} is common across blockholders while $(\sigma^2_{\varepsilon_b}, k_b)$ are blockholder-specific, the cross-blockholder voting covariances $\mathrm{E}[V_{bj} \cdot V_{b'j}]$ identify σ^2_{uB}. Conditional on (τ_j, S_j), blockholders co-vote only through their shared exposure to the residual common factor η_{Bj}, and the strength of this co-voting is governed by σ^2_{uB}. Once σ^2_{uB} is pinned down, each blockholder's $(\sigma^2_{\varepsilon_b}, k_b)$ is obtained by inversion from the individual probit coefficients $(\alpha_{b0}, \alpha_{b1}, \alpha_{b2})$.

The importance of cross-blockholder covariances. The three moments from each blockholder probit do not, when aggregated, identify σ^2_{uB}. Intuitively, Corollary 1 (stated below) shows that we could change σ^2_{uB} and compensate by adjusting $\sigma^2_{\varepsilon b}$ and k_b across blockholders to generate the same voting probabilities. The covariances break this because they measure co-movement in votes after conditioning on (τ_j, S_j). The model's identifying assumption for σ^2_{uB} is that the only remaining source of co-movement is shared exposure to η_{Bj}.

4.1.3. Preferences

Finally, we identify the preference parameters

$$\Delta = \left\{\, \delta_D, \{\delta_b\}_{b=1}^{N_B} \right\}$$

from the best-response conditions at the cutoffs. For dispersed shareholders,

$$E\big[x_j \mid m_{Dj} = k_D,\ \mathrm{piv}_{Dj}\big] + \delta_D = 0,$$

and the argument is analogous for each blockholder b. Once the information environment is fully pinned down, the conditional expectation $E[x_j \mid m_{ij} = k_i, \mathrm{piv}_{ij}]$ is a known function of the information parameters, so each preference δ_i is uniquely determined by the corresponding indifference condition.

4.2. Formal Identification

We formalize the identification claim with the following proposition.

Proposition 1: Identification of model parameters

Let the structural parameter vector be

$$\Theta \equiv \left(\rho, \xi, \sigma^2_{uD}, \sigma^2_{uB}, \sigma^2_{\varepsilon D}, \{\sigma^2_{\varepsilon_b}\}_{b=1}^{N_B}, k_D, \{k_b\}_{b=1}^{N_B}, \delta_D, \{\delta_b\}_{b=1}^{N_B}\right),$$

Given the joint cross-sectional distribution of

$$\left\{\tau_j, S_j, \{V_{bj}\}_{b=1}^{N_B}\right\}$$

across proposals (together with the ownership weights $\{\psi_b\}_{b=1}^{B}$ and the model's voting rule), the parameter vector Θ is uniquely identified.

Proof. See Appendix B.1. ■

4.2.1. Implications

Proposition 1 establishes identification, which relies on two key ingredients. First, a structural model of the information environment that explains how public and private information influence each agent's beliefs. Second, conditional on our structural model, cross-blockholder voting covariances conditional on (τ_j, S_j), which pin down the common information precision. A natural question is whether the full joint distribution is necessary, or whether the marginal distributions of individual voting patterns suffice. The following corollary shows that they do not, even under a common information environment in which all shareholders load equally on a single public signal.[14] We then show that this result has important implications for the empirical literature studying information and preferences in shareholder voting.

Corollary 1: Non-Identification from Individual Voting Patterns

[14]This nests the full model as the case $\sigma^2_{u_D} = \sigma^2_{u_B}$. All other primitives are unchanged.

Specialize the model in Section 2 to $\sigma^2_{uD} = \sigma^2_{uB} = \sigma^2_u$.[15] *Let*

$$\Gamma = \left(\sigma^2_u, \sigma^2_{\varepsilon_D}, \{\sigma^2_{\varepsilon_b}\}_{b=1}^{N_B}, \delta_D, \{\delta_b\}_{b=1}^{N_B}\right)$$

be a set of parameters supporting an equilibrium with cutoff profile $\mathbf{k} = (k_D, \{k_b\}_b)$ and generating observed marginal distributions of (τ_j, V_{bj}). For each Γ, there exists a $\tilde{\Gamma}$, indexed by $\tilde{\sigma}^2_u$ near σ^2_u, that supports an equilibrium with the same marginal distributions of (τ_j, V_{bj}).

Proof. See Appendix B.2. ■

Intuitively, the corollary states that one cannot use observed individual voting patterns to separately identify information from preferences. The source of non-identification is that σ^2_u, the precision of public information, is a free parameter when only marginal distributions are observed. Changing σ^2_u alters the weight shareholders place on public relative to private information, and simultaneously changes the informativeness of the pivotal event. An investor who appears to support management may be unbiased but strategically responsive to pivotality under one information environment, or actually biased under another. Individual voting patterns cannot separate the two.

What breaks this observational equivalence, and delivers the identification in Proposition 1, is the covariance of votes across different blockholders, conditional on (τ_j, S_j). These joint moments pin down the common information precision, which is precisely the channel that marginal distributions leave unidentified.

The corollary has implications for empirical approaches that use voting data to characterize either investor preferences or information processing while maintaining assumptions about the other. Because observed voting behavior reflects both channels simultaneously, estimates from either approach are sensitive to these maintained assumptions. The corollary establishes the existence of observationally equivalent parameters but is silent on their quantitative distance in any particular application.

[15] To clarify, we do not assume that each group sees the same public signal, we are simply reducing the number of parameters to isolate the source of non-identification.

Conditional means. Corollary 1 implies that an empirical strategy relying only on mean support rates by investor type cannot separately identify information and preferences. For example, consider a standard reduced-form regression estimated using a panel of (block) voters and proposals across firms:

$$\text{Vote}_{ij} = \mu_i + \mu_j + \beta \cdot \text{ISS}_j + \gamma' X_{ij} + \epsilon_{ij}.$$

This specification uses only the conditional mean of each investor's vote given observables. The investor fixed effect μ_i is affected by how precisely investor i processes public information relative to private information, the strategic effect of pivotality conditional on the information environment, and her innate preference δ_i. An investor with a high μ_i may be pro-management, or may be unbiased but strategically responsive to pivotality in an information environment where being decisive is good news about the proposal. Symmetrically, interpreting μ_i as a measure of information quality requires maintaining assumptions about preferences. Including ISS_j as a control absorbs the average effect of the public signal but does not fully resolve this, since μ_i still reflects investor i's reliance on public relative to private information.

Covariance structure. A recent literature applies dimensionality reduction methods to the observed voting matrix to extract low-dimensional spatial representations of investor behavior (Bubb and Catan, 2022; Bolton et al., 2020). These approaches use cross-investor covariances, but a similar identification challenge arises: even shareholders with identical preferences co-vote through shared public signals and through conditioning on the pivotal event.[16] Without explicit structure on the information environment, the estimated spatial coordinates reflect a combination of preference similarity, shared public information, and equilibrium responses to pivotality. Our structural model provides the additional restrictions needed to isolate these channels.

[16] Variation in public signal precision σ_u^2 changes the weight shareholders place on common signals relative to private information, generating proposal-level covariance that mimics preference similarity.

4.3. Estimation Routine

We take the moment restrictions implied by the identification argument above and stack them into a single sample moment vector. At the proposal level, we match the conditional means $E[T_j \mid S_j]$ and conditional variances $\mathrm{Var}(T_j \mid S_j)$ for $S_j \in \{0, 1\}$, together with the probit intercept and slope from regressing S_j on T_j. At the blockholder level, we match score moments implied by each blockholder's vote probit: $E[V_{bj}]$, $E[V_{bj}T_j]$, $E[V_{bj}S_j]$, and $E[V_{bj}T_jS_j]$, where the last provides an overidentifying restriction. We also match cross-blockholder co-voting moments $E[V_{bj} \cdot V_{b'j}]$ for all blockholder pairs, which discipline the common public-signal variance σ^2_{uB}. Estimation then proceeds via a standard two-step GMM routine: we obtain a first-step estimate using an initial weighting matrix, form the optimal weighting matrix from the estimated moment covariance, and re-estimate in a second step. Appendix B.3 details in full.

4.4. Data

To estimate the model, we focus on executive compensation proposals ("Say-on-Pay"), in which shareholders cast advisory votes on management's disclosed compensation program. These are management-sponsored proposals, so a "For" vote is naturally interpreted as support for management. Because SOP is advisory, there is no statutory voting threshold that compels the compensation committee to revise pay. Instead, what matters is the level of dissent at which key governance intermediaries and boards treat the outcome as a negative signal. A central benchmark is 70% support. In particular, Institutional Shareholder Services (ISS) treats SOP support below 70% as a low-support outcome that triggers heightened scrutiny of the firm's response and may lead ISS to recommend votes against the subsequent SOP proposal and/or directors (ISS, 2022, Section 5, "Compensation").[17] For these reasons, we use 70% as the passing threshold in the empirical analyses.

[17]Consistent with this, firms experience sharp changes in institutional investor engagement around the 70% cutoff, while placebo threshold analyses do not show comparable discontinuities at alternative cutoffs such as 50% support (Dey et al., 2024). More broadly, low SOP support is widely viewed as a signal of shareholder dissatisfaction and can trigger engagement and compensation changes (Ertimur et al., 2013). Moreover, Barry (2026) shows that 70% support is an important threshold in influencing compensation policy.

4.4.1. Sample Construction

We use the ISS Voting Results dataset to gather a universe of Say-on-Pay proposals held at annual meetings in 2020. We identify Say-on-Pay votes using the ISS Agenda Item ID "M0550." For each proposal, we collect identifying information about the firm, the date of the meeting, the number of shares voted for/Against/abstain, ISS's voting recommendation, and additional variables related to voting procedures. We keep only proposals with voting bases "For + Against + Abstain" or "For + Against." We merge share price information from the CRSP monthly stock file, matching on the month-end prior to the meeting date. We compute the proposal's total support rate, λ_j, following:

$$\lambda_j = \begin{cases} \dfrac{\#\text{ shares for}}{\#\text{ shares for} + \#\text{ shares Against}}, & \text{if base = "For + Against,"} \\ \dfrac{\#\text{ shares for}}{\#\text{ shares for} + \#\text{ shares Against} + \#\text{ shares abstain}}, & \text{if base = "For + Against + Abstain".} \end{cases}$$

We follow the approach in Bubb and Catan (2022) and Bolton et al. (2020) and filter to proposals receiving less than 95% support. We also compute turnout, χ_j:

$$\chi_j = \begin{cases} \dfrac{\#\text{ shares for} + \#\text{ shares Against}}{\#\text{ shares outstanding}}, & \text{if base = "For + Against,"} \\ \dfrac{\#\text{ shares for} + \#\text{ shares Against} + \#\text{ shares abstain}}{\#\text{ shares outstanding}}, & \text{if base = "For + Against + Abstain".} \end{cases}$$

We drop proposals with less than 50% turnout. For each Say-on-Pay proposal, we download all mutual fund voting records. We collect the name and ISS identifier of the mutual fund, the mutual fund family, and the vote cast. We keep only for/Against/abstain votes. We aggregate fund-level votes to the family-level, deeming a family to vote for the proposal if a majority of its funds support it. We download institutional ownership data from the Thomson Reuters 13F dataset, merging on the quarter prior to the meeting date. We compute each manager's ownership by dividing the number of shares held by the number of shares outstanding.

We now turn to how we map the raw data to the structure assumed in the model. First, we compute each 13F manager's turnout-adjusted ownership by dividing their raw ownership by the turnout rate χ_j. We deem a manager to be a blockholder if its turnout-adjusted ownership is at least five percent. Next, we match 13F manager names to ISS fund family names to record each blockholder's vote.[18] For each proposal, this step produces a set of blockholders, their votes, and their turnout-adjusted ownership. We construct the vector of blockholder ownership, ψ^{χ}_{Bj}, and the vector of blockholder votes, V^B_j, where the number of blockholders N_B is the count of unique blockholders across all proposals. Given the total support rate, λ_j, we impute the dispersed shareholders' support rate, τ_j, following

$$\tau_j = \frac{\lambda_j - V_j^{B\top} \psi^{\chi}_{Bj}}{\Psi^{\chi}_{Dj}}.$$

Here, Ψ^{χ}_{Dj} denotes the dispersed shareholders' turnout-adjusted ownership share in proposal j. This step produces a series of voting results $\{\tau_j, V^B_j\}$ and ownership structures $\{\Psi_j\}$ suitable as inputs to the procedure described in Section 4.

4.4.2. Sample summary

We summarize our sample of 993 Say-on-Pay votes in Table 1. The average firm has 2.89 blockholders, with combined ownership of 29.53%. Average total support is 83.65%. Support is lower among dispersed shareholders, with an average dispersed support rate of 81.56%, while blockholders are more supportive on average: conditional on being a blockholder, the mean blockholder support rate is 86.92%. Using the standard convention that votes with less than 70% support are considered failures, 85.90% of proposals pass.

Panel B summarizes ownership and voting behavior for the largest blockholders. Vanguard is the most frequent blockholder, appearing in 909 votes, while BlackRock appears in 800. State Street appears in 291 votes, consistent with its smaller average stake. The Big Three hold economically large positions: average ownership is 10.65% for Vanguard, 11.62% for BlackRock, and 6.63% for State Street. All three are more supportive than dispersed shareholders on average: Vanguard supports 90.21% of proposals, BlackRock supports 95.00%, and State Street supports 90.38%. Pivot probabilities vary meaningfully across investors, reflecting

[18] In the underlying data, we keep only for/Against/abstain votes; fund-level votes are aggregated to the family level.

both differences in stake sizes and voting behavior: the average pivot probability is 12.54% for Vanguard and 17.00% for BlackRock, compared to 4.12% for State Street.

5. Results

5.1. Model fit

Overall, the model and data moments closely align. Figure 3 illustrates the GMM estimation results by scattering model moments against data moments. The figure shows that all moments lie close to the 45-degree line, suggesting that the model can match the data.

Table 2 displays model fit on important moments. At the proposal-level (Panel A), the table shows that we match total support. It also shows that the proportion of passing and close votes is very similar to the data. In Panel B, we report each blockholder's voting probability conditional on the ISS recommendation, along with its estimated pivot probability (i.e., the share of proposals for which switching that blockholder's vote would change the outcome). Overall, the model matches these blockholder-level moments closely.

One notable deviation is that the model assigns a higher pivot probability to Dimensional in the model. This discrepancy is economically interpretable: we estimate that Dimensional has the strongest anti-management preference among blockholders, which makes its vote most likely to be decisive in proposals that generate pivotal events. In other words, the model's tendency to overstate Dimensional's pivotality reflects an implication of the preference estimates, when an investor is systematically more willing to oppose management, it endogenously becomes pivotal more often.

Figure 4 displays simulated total support from the model as compared to the data. We display the densities of support conditional on the ISS recommendation. The figure shows that the model is able to closely capture the data generating process for support rates. However, the model cannot perfectly replicate the peakedness of support in the data around the mean.

5.2. Parameter Estimates

Table 3 reports the estimated parameters. The proxy advisor signal is aligned with the information of dispersed shareholders: the estimated correlation is $\rho = 0.751$. This implies that the advisor recommendation is not orthogonal, but rather a noisy public summary that loads heavily on the same underlying component of proposal quality that dispersed investors would infer from their own information. Put differently, shifts in the recommendation are informative because they move in tandem with what the broader investor base believes.

Second, the model estimates meaningful heterogeneity in public information precision across voter types. Dispersed shareholders' public signal is relatively noisy ($\sigma_{u_D} = 6.743$), whereas blockholders observe a more precise public signal ($\sigma_{u_B} = 2.537$).

Turning to private information, the pattern goes in the opposite direction: blockholders' private signals are substantially more precise than those of dispersed shareholders. Dispersed shareholders have $\sigma_D = 2.612$, while most blockholders have much lower private signal volatility. For example, among blockholders, Vanguard has the most precise at 0.468 and T. Rowe the most at 1.579.

Finally, the estimated preference parameters δ_i shift a shareholder's ideal point toward passage. $\Phi(\delta_i)$ summarizes this on a probability scale: it is the share of proposals that shareholder i would ideally like to see pass, given the prior distribution of proposal quality and absent any information or strategic considerations. Because $\Phi(\delta_i)$ depends on the assumed state distribution, its level should be interpreted with caution; however, cross-sectional comparisons across shareholders are informative about relative preferences.[19] Dispersed shareholders have $\Phi(\delta_D) = 81.52\%$. Most large blockholders are more pro-passage: Vanguard (88.45%), BlackRock (90.35%), State Street (87.74%), Fidelity (85.75%), and T. Rowe (81.94%). Dimensional is a clear outlier at 64.91%, the only blockholder less pro-passage than the dispersed base.

Importantly, these preference indices are not the same object as raw voting frequencies: Table 1 shows that vote-for rates are substantially higher for several investors (e.g., Vanguard 90.2% and BlackRock 95.0%) than their corresponding $\Phi(\delta_i)$ values, underscoring that equilibrium support need not move one-for-one with underlying preferences.

5.3. Decomposing Voting: Strategic Incentives and Information

Maug and Rydqvist (2009) argue that strategic voting is essential for shareholder voting to

[19] This quantity is a preference index and should not be compared to the 70% support threshold used to classify Say-on-Pay outcomes.

screen proposals effectively: by conditioning on the pivotal event, shareholders aggregate the information of others into their own decision, and without this channel, voting based on noisy private signals alone may fail to screen out low-quality proposals. To quantify this mechanism, we compare the estimated equilibrium to a non-strategic counterfactual that holds fixed the information environment but removes the effect of pivotality on voting decisions.

In the baseline model, shareholder i votes

$$V_{ij} = \mathbf{1}\{ m_{ij} \geq k_i \},$$

where k_i is pinned down by $E[x_j \mid \eta_j, m_{ij} = k_i, \text{piv}_{ij}] = -\delta_i$. In the non-strategic counterfactual, we instead set the cutoff to $-\delta_i$ directly:

$$V_{ij}^{NS} = \mathbf{1}\{ m_{ij} \geq -\delta_i \}.$$

The only difference is that shareholders no longer condition on the event of being pivotal.

Table 4 reports the impact on support probabilities by shareholder type.Removing pivotality increases support across most types, with the largest effects where pivotality is most consequential for the marginal decision: dispersed shareholders' support rises from 81.56% to 98.93%. This is the swing voter's curse in action (Feddersen and Pesendorfer, 1996). Because dispersed shareholders are less informed and less biased than blockholders, a dispersed shareholder who conditions on being pivotal rationally infers that proposal quality is relatively low: if the state were sufficiently good, the better-informed blockholders would already have pushed the proposal safely above the threshold. Strategic voting therefore makes dispersed shareholders substantially more conservative than under the non-strategic benchmark, and this effect is larger for them than for other shareholder types.[20]

[20] Formally, for any posterior mean m,

$$\mathrm{E}[x_j \mid \eta_{Dj}, m_{Dj} = m, \text{piv}_{Dj}] = \frac{\int x \Pr\big(\text{piv}_{Dj} \mid x, \eta_{Dj}, m\big) f(x \mid \eta_{Dj}, m\big) \; dx}{\int \Pr\big(\text{piv}_{Dj} \mid x, \eta_{Dj}, m\big) f(x \mid \eta_{Dj}, m)\big) \; dx}.$$

In the continuum limit, conditioning on pivotality concentrates the posterior on the unique state x^* at which the aggregate vote in favor equals the passing threshold; pivotality thus acts as a common public signal for dispersed that dominates their own information. Because blockholders are more strongly pro-passage and better informed, they supply most of the votes needed for passage on their own, so x^* lies below what dispersed

For other blockholders, strategic voting acts as a screening device (Maug, 1999) that provides them with a signal about the proposal's underlying quality. Dispersed shareholders vote conservatively, so the marginal vote arises only when the proposal is good enough to draw meaningful dispersed support. Pivotality thus shifts a blockholder's posterior toward higher x, pulling her cutoff below the non-strategic benchmark. Columns (2)–(7) of Panel A display this: every institutional blockholder except Dimensional supports proposals more often under the strategic equilibrium than under the non-strategic benchmark.[21]

5.4. Information and Preference Heterogeneity in Shareholder Voting

A central question in the governance literature is whether blockholders vote differently from other shareholders because of differences in preferences or in information (Hu et al., 2024; Matsusaka and Shu, 2022; Bubb and Catan, 2022; Bolton et al., 2020). We implement counterfactuals that separately equalize information and preferences across investor types by assigning blockholders the same public and private information precision as dispersed shareholders, and separately the same preferences, re-solving for the equilibrium under each counterfactual set of parameters.

Table 5 reports counterfactual voting outcomes when blockholders are assigned dispersed preferences (column 2), dispersed information (column 3), or both (column 4).

Equalizing preferences pulls dispersed and blockholder support toward a common level: dispersed support rises from 81.56% to 83.76% and blockholder support falls from 87.21% to 83.21%. The close vote rate edges down to 1 and total support is essentially unchanged. The close vote rate, which represents shareholders' ability to effectively separate good proposals from bad, falls slightly, but is largely unchanged.

Equalizing information, which removes the innate information advantage blockholders have, has a much larger impact. While support rates are somewhat similar to the baseline, the

shareholders would accept under their own preferences. Hence

$$\mathrm{E}[x_j \mid \eta_{Dj}, m_{Dj} = m, \mathrm{piv}_{Dj}] \ll \mathrm{E}[x_j \mid \eta_{Dj}, m_{Dj} = m].$$

[21] Dimensional is the only blockholder whose preference lies meaningfully below that of dispersed shareholders, so the screening logic runs in reverse: pivotality conveys that the proposal is good enough to clear the dispersed bar, which still leaves it below Dimensional's, and the strategic cutoff rises rather than falls.

close vote rate jumps from 15.11% to 26.08%, and the cross-proposal variance of total support collapses from 2.04% to 1.36%.

The close vote rate and the variance of total support tell the same story. Blockholders' precise signals make their vote shares track proposal quality, pushing both bad and good proposals away from the threshold. Removing that state-contingency makes voting outcomes cluster near the threshold regardless of the value of x_j. Equalizing preferences, on the other hand leaves the state-contingency intact and has a small impact on aggregate outcomes.

While our results suggest that blockholders are somewhat biased toward management relative to dispersed shareholders, they vote in favor because their precise information tells them a proposal is good or bad, even when it is marginally close to the threshold. This has important implications for the governance literature and policy. Policies targeted at better representation of preferences (e.g., pass-through voting) risk distorting the informational advantage of blockholders.

6. Conclusion

Institutional investors are central participants in shareholder voting, yet observed vote records may not map into underlying preferences when proposals are evaluated under incomplete information and voters act strategically. This paper develops and estimates a structural model of management-sponsored proposals in which shareholders differ in ownership stakes, information precision, and intrinsic preferences for passage. In equilibrium, voting follows cutoff strategies that reflect not only investors' own signals, but also strategic inference from the voting environment itself, particularly the information embedded in being pivotal.

Estimating the model using Say-on-Pay votes and mutual fund voting records, we recover institution-level information and preference parameters and show that preference-implied support can diverge sharply from realized support. Large asset managers that appear uniformly pro-management in raw voting data exhibit much more moderate baseline preferences once correlated information and pivotality incentives are accounted for. At the same time, the estimates imply substantial heterogeneity in information: large institutions operate with meaningfully more precise effective public information and private assessments than dispersed shareholders, so higher observed support need not reflect weaker monitoring or a stronger taste for passage.

Counterfactuals that remove pivotality-driven incentives while holding preferences and the information environment fixed highlight the first-order role of strategic voting in shaping outcomes. Strategic conditioning makes voting more conservative in contentious states, lowering overall support and increasing the incidence of close votes. In the estimation sample, eliminating strategic voting raises the passing rate from about 85% to nearly 96%, while cutting the close-vote rate roughly in half. Taken together, the results imply that interpreting shareholder voice requires disentangling information from preferences and recognizing that the voting environment itself materially shapes voting outcomes.

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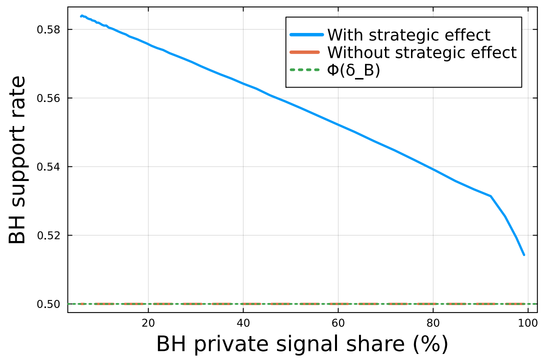
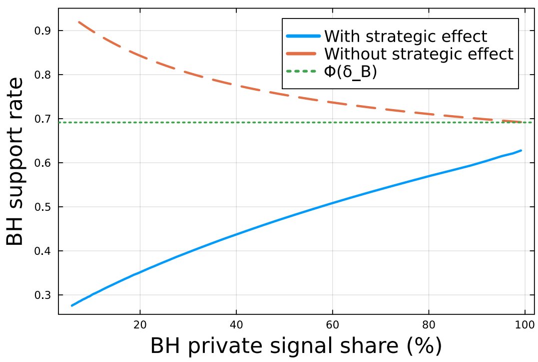
Figure 1. Comparative static exercise: Signal and strategic effects

This figure presents the comparative static exercise from Section 3.1. We assume one blockholder that owns 20% of the firm, with the remaining 80% held by dispersed shareholders. For this comparative static exercise, we maintain the following parameters constant as we vary δ_b and $\sigma_{\epsilon b}$: $\sigma_{uD} = \sigma_{uB} = 4$, $\sigma_{\epsilon D} = 1$, $\sigma_{\epsilon b} = 10$, $\lambda^* = 0.7$. For each level of the blockholder's private-signal precision, holding the other parameters fixed, the equilibrium cutoffs of the dispersed shareholders and the blockholder are solved under strategic voting. Three objects are then computed: the preference-implied support rate, $\Phi(\delta_B)$, which reflects only the blockholder's bias; the non-strategic support rate, which is the support probability under the same information structure but without strategic voting, so that the blockholder votes according to $m_b \geq -\delta_b$; and the strategic support rate, which is the equilibrium support probability under the strategic voting rule $m_b \geq k_b$. The three support rates are plotted against the share of the blockholder's private signal precision relative to the total precision contributed by the prior, public signal, and private signal, i.e., $\frac{1/\sigma^2_{\epsilon b}}{1+1/\sigma^2_{\epsilon b}+1/\sigma^2_{ub}}$. Comparing these three support rates isolates the strategic and signal effects.

(a) $\delta_b = 0$



(b) $\delta_b = 0.5$



(c) $\delta_b = -0.5$

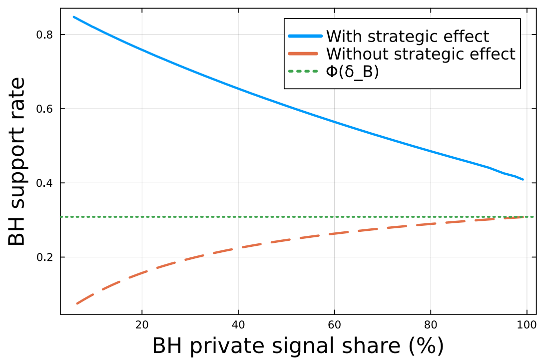


Figure 2. Comparative static exercise: Observationally equivalent blockholder support rate under different information and preference parameters

This figure presents the comparative static exercise from Section 3.2. We fix the support rate and vary σ_b and δ_b to back out the same observed support rate, holding other parameters fixed: $\sigma_{uD} = \sigma_{uB} = 1, \sigma_{\epsilon D} = 1, \lambda^* = 0.5$. The figure plots the preference-implied support rates $\Phi(\delta_b)$ versus the blockholder's private signal precision share with and without strategic effects. All environments produce 75% observed blockholder support (horizontal dashed line) and 50% observed dispersed shareholder support rate. The solid curve plots the preference-implied support rate that is required to sustain the same 75% support rate taking into consideration the strategic effect, while the dashed curve shuts down the strategic effect.

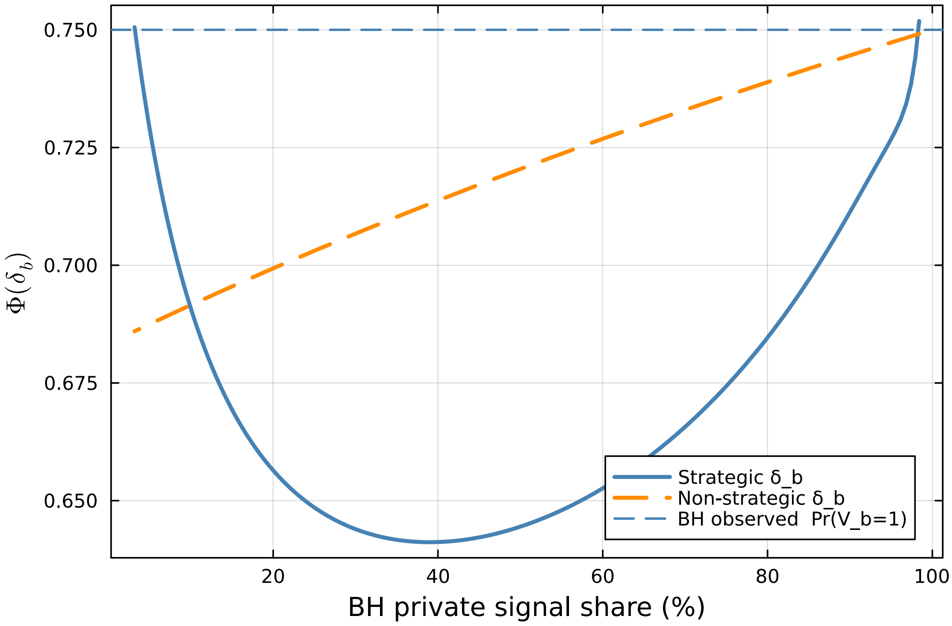


Figure 3. Model fit: Closeness in data and model moments

This figure shows the closeness in data and model moments. Each point plots a data moment against the corresponding model-predicted moment, with the dashed line indicating perfect fit. Moments are grouped into dispersed/advisor (circles), blockholder-specific (triangles), and cross-blockholder correlations (diamonds). Both axes use a symmetric-log transform, $\operatorname{sgn}(x)\log(1 + |x|)$. The correlation between data and model moments exceeds 0.99.

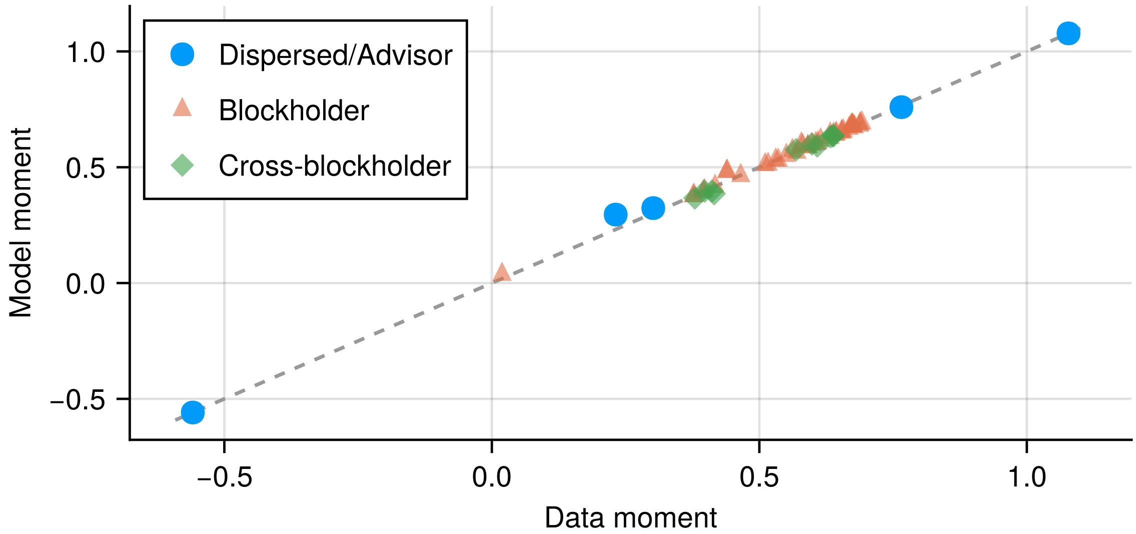


Figure 4. Model fit: Distribution of support

This figure compares the empirical distribution of support rates to the distribution implied by simulations of the estimated model. The solid lines plot kernel density estimates of the simulated distributions and the dashed lines plot the same from the data. To construct the simulated distributions, we hold each proposal's blockholder ownership structure fixed at its empirical values (the set of blockholders, their ownership shares $\{\psi_{bj}\}_b$, and associated blockholder counts), and draw the latent state and shocks from the estimated model. Given simulated dispersed support and ISS recommendation(τ_j, S_j), we generate blockholder support using the blockholder-level model-implied voting probabilities; total support is then $\tau_j + s_{Bj}$.

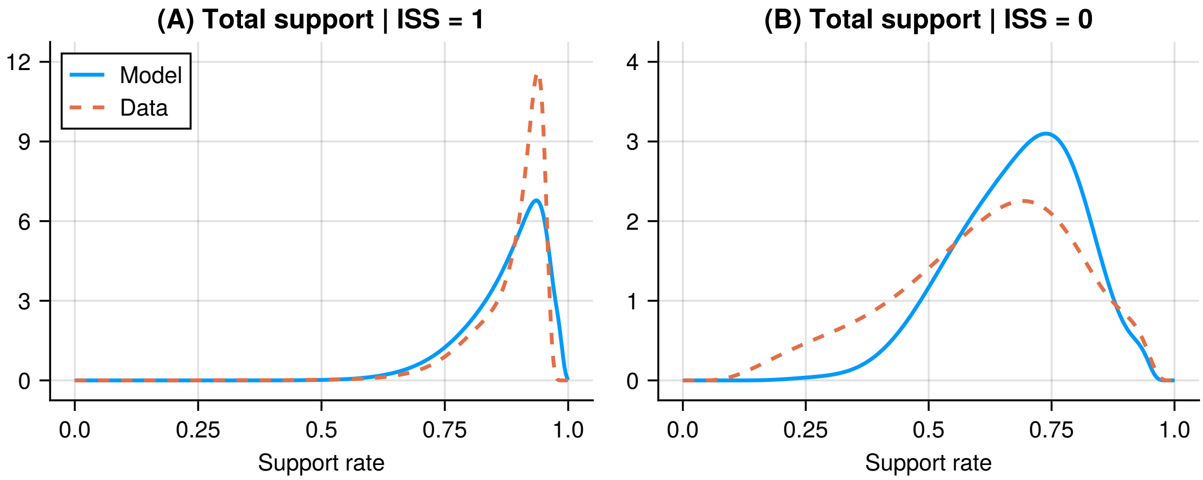


Table 1. Summary statistics

This table summarizes the sample of Say-on-Pay proposals we use to estimate the model. In Panel A, we provide statistics at the proposal-level. Number of blockholders is the count of blockholders that own the firm and whom we can match a vote to; we define a blockholder as a shareholder who owns more than five percent of the firm's shares on a turnout-adjusted basis. Total support is the number of shares voted in support of the proposal divided by the number of votes cast. Dispersed and blockholder support are the percent shareholders supporting the proposal, conditional on type, imputed using the method described in the text. Passing vote is a dummy variable equal to one if the proposal's support rate exceeds the 70% passing threshold. Close vote a dummy variable equal to one if the proposal's support rate is between 65% and 75%. In Panel B, we provide statistics at the blockholder-level. Number of votes lists the number of proposals in the sample that the blockholder appears as a block shareholder. Average ownership is the blockholder's average ownership across all proposals. Support rate is the percent of proposals supported by the blockholder. Pivot probability is the percent of proposals where had the blockholder changed its vote, the proposal would have passed (failed) if it actually failed (passed).

Panel A. Proposal-level

	(1)	(2)	(3)	(4)	(5)	(6)
	N	Mean	SD	25th	50th	75th
Number of blockholders	993	2.89	1.08	2	3	4
Blockholder ownership	993	29.53%	12.86%	20.16%	28.92%	38.27%
Total support	993	83.65%	14.11%	79.67%	89.53%	93.24%
Blockholder support	993	86.92%	25.65%	82.58%	100.00%	100.00%
Dispersed support	993	81.56%	14.40%	77.46%	87.12%	90.86%
Passing vote	993	85.90%				
Close vote	993	8.86%				

Panel B. Blockholder-level

	(1)	(2)	(3)	(4)	(5)	(6)
	Vanguard	BlackRock	State Street	Dimensional	Fidelity	T. Rowe
Number of votes	909	800	291	285	130	118
Average ownership	10.65%	11.62%	6.63%	8.49%	9.65%	9.83%
Support rate	90.21%	95.00%	90.38%	48.77%	83.85%	92.37%
Pivot probability	12.54%	17.00%	4.12%	7.02%	16.92%	11.02%

Table 2. Model fit

This table summarizes the fit of the structural voting model estimated by GMM. Panel A reports dispersed-side moments based on the joint distribution of the inverse–probit index $T = \Phi^{-1}(\tau)$ and the ISS recommendation S: the conditional means and variances $\mathrm{E}[T \mid S = x]$ and $\mathrm{Var}(T \mid S = x)$ for $x \in 0, 1$, and the probit slope β_T^{ISS} from regressing S on T. Panel B reports blockholder moments used in estimation. For each blockholder b, we report the moments $\mathrm{E}[V_b]$, $\mathrm{E}[V_b \cdot T]$, and $\mathrm{E}[V_b \cdot S]$ (and their model-implied counterparts), which capture the level of support, the comovement of blockholder voting with dispersed support, and the incremental shift associated with the ISS recommendation. These moments are the GMM analogs of an auxiliary probit representation of blockholder voting in (T, S).

Panel A. Proposal-level

	(1)	(2)
	Data	Model
$\mathbb{E}$[Dispersed support | ISS = 0]	61.82%	67.58%
Var(Dispersed support | ISS = 0)	3.26%	1.74%
$\mathbb{E}$[Dispersed support | ISS = 1]	86.11%	86.12%
Var(Dispersed support | ISS = 1)	0.70%	0.64%
$\mathbb{E}$[Total support | ISS = 0]	61.79%	66.15%
Var(Total support | ISS = 0)	2.65%	1.46%
$\mathbb{E}$[Total support | ISS = 1]	88.69%	88.40%
Var(Total support | ISS = 1)	0.49%	0.42%
Passing vote	85.90%	86.39%
Close vote	18.03%	20.46%
Pr(ISS = 1)	81.27%	81.25%
Probit AME of Φ^{-1}(Dispersed support) on Pr(ISS = 1)	1.28%	1.48%

Panel B. Blockholder-level

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Vanguard		BlackRock		State Street		Dimensional		Fidelity		T. Rowe	
	Data	Model	Data	Model	Data	Model	Data	Model	Data	Model	Data	Model
$\Pr(V_b = 1 \mid \text{ISS} = 0)$	51.76%	51.77%	73.43%	73.43%	45.71%	45.72%	1.92%	3.80%	45.95%	45.94%	70.00%	70.01%
$\Pr(V_b = 1 \mid \text{ISS} = 1)$	99.05%	98.86%	99.70%	99.64%	96.48%	97.17%	59.23%	59.00%	98.92%	97.70%	96.94%	96.24%
Pivot probability	12.54%	16.01%	17.00%	24.37%	4.12%	5.21%	7.02%	9.02%	16.92%	10.62%	11.02%	13.47%

Table 3. Estimated parameters

This table reports the estimated parameters of the information structure and voting behavior. Column 1 displays parameters relating to the public information environment: ξ and ρ describe the advisor threshold and the correlation of the latent advisor signal with the latent dispersed shareholder signal. σ_{u_D} and σ_{u_B} represent the idiosyncratic volatility of the (combined) public signals for dispersed and block shareholders respectively. Column 2-8 display parameters relating to each distinct shareholder type in the economy: $i \in \left\{D, \{b\}_1^B\right\}$. For example, column 2 displays dispersed shareholder parameters: the idiosyncratic volatility of the dispersed private signal σ_D, the equilibrium dispersed voting threshold k_D and the preference δ_D. In the last row, we also display each shareholder's preference-implied support rate $\Phi(\delta_i)$, which tells us, given the prior, the proportion of votes each shareholder would prefer to pass. Standard errors, clustered at the proposal level and estimated via bootstrap, are reported in parentheses; see Appendix B.3 for details.

Panel A. Parameter values

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
			Blockholders					
Parameter	Public information	Dispersed	Vanguard	BlackRock	State Street	Dimensional	Fidelity	T. Rowe
ξ	-0.739							
	(0.002)							
ρ	0.751							
	(0.001)							
σ_{u_D}	6.744							
	(0.009)							
σ_{u_B}	2.562							
	(0.861)							
σ_i		2.612	0.467	0.563	0.719	0.740	0.523	1.574
		(0.005)	(0.006)	(0.011)	(0.022)	(0.023)	(0.010)	(0.200)
k_i		-0.316	-1.129	-1.404	-0.957	0.107	-1.010	-0.819
		(0.000)	(0.001)	(0.003)	(0.003)	(0.001)	(0.005)	(0.004)
δ_i		0.897	1.197	1.300	1.162	0.383	1.068	0.917
		(0.002)	(0.006)	(0.023)	(0.029)	(0.017)	(0.015)	(0.023)

Panel B. Parameter interpretations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Blockholders					
Parameter	Dispersed	Vanguard	BlackRock	State Street	Dimensional	Fidelity	T. Rowe
Preference-implied support rates							
$\Phi(\delta_i)$	81.52%	88.44%	90.33%	87.73%	64.93%	85.73%	82.05%
	(0.04%)	(0.11%)	(0.39%)	(0.59%)	(0.64%)	(0.33%)	(0.61%)
Variance shares							
Prior	85.57%	17.44%	23.24%	32.38%	33.57%	20.82%	64.26%
	(0.04%)	(0.49%)	(0.88%)	(1.63%)	(1.72%)	(0.73%)	(5.99%)
Common signal	1.88%	2.66%	3.54%	4.93%	5.12%	3.17%	9.79%
	(0.01%)	(1.74%)	(2.30%)	(3.16%)	(3.27%)	(2.07%)	(5.97%)
Private signal	12.54%	79.90%	73.22%	62.69%	61.32%	76.01%	25.95%
	(0.05%)	(1.49%)	(1.91%)	(2.52%)	(2.57%)	(1.75%)	(5.18%)

Table 4. The impact of strategic voting on shareholder-level support

This table compares shareholder-level voting probabilities under the estimated equilibrium and a non-strategic counterfactual in which each shareholder votes $V_{ij}^{NS} = \mathbf{1}\{m_{ij} \geq -\delta_i\}$, removing the effect of pivotality on voting decisions while holding fixed the information environment. Panel A reports results for all proposals. Panel B splits by ISS recommendation.

Panel A. All proposals

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dispersed	Vanguard	BlackRock	State Street	Dimensional	Fidelity	T. Rowe
Data	81.56%	90.21%	95.00%	90.38%	48.77%	83.85%	92.37%
Full model	81.56%	90.05%	94.95%	90.98%	48.93%	82.97%	91.79%
Non-strategic	98.93%	84.34%	87.23%	88.47%	54.57%	74.67%	88.37%

Panel B. By ISS recommendation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	Dispersed		Vanguard		BlackRock		State Street		Dimensional		Fidelity		T. Rowe	
	ISS=0	ISS=1	ISS=0	ISS=1	ISS=0	ISS=1	ISS=0	ISS=1	ISS=0	ISS=1	ISS=0	ISS=1	ISS=0	ISS=1
Data	61.82%	86.11%	51.76%	99.05%	73.43%	99.70%	45.71%	96.48%	1.92%	59.23%	45.95%	98.92%	70.00%	96.94%
Full model	61.82%	86.11%	51.77%	98.86%	73.43%	99.64%	45.72%	97.17%	3.80%	59.00%	45.94%	97.70%	70.01%	96.24%
Non-strategic	96.21%	99.56%	26.24%	97.71%	35.99%	98.38%	28.75%	96.64%	7.97%	64.97%	22.65%	95.37%	56.25%	94.93%

Table 5. The impact of information and preferences on shareholder voting outcomes

This table displays the impact of equalizing preferences and information across shareholders. In particular, we first set all blockholders to have the same preference as dispersed shareholders (i.e., $\delta_b = \delta_D$ for all b). Then, we set all blockholders to have the same information quality as dispersed shareholders (i.e., $\sigma_b = \sigma_D$ for all b and $\sigma_{uB} = \sigma_{uD}$.) To estimate each counterfactual, we solve the model under the counterfactual parameter vector, and then use the moment-generating conditions to build counterfactual outcomes. In Panel A, in column (1), for consistency, we also solve the model under the estimated parameters, so the displayed moments are not perfectly consistent with the GMM-based moments (though very close). In column 2, we display results from the equal preferences counterfactal, and in column 3 we display results from the equal information counterfactual. Column 4 displays results from a counterfactual in which both preferences and information are equalized. Panel B displays analagous, split by ISS recommendation.

Panel A.

	(1)	(2)	(3)	(4)
	Full model	Equal prefs	Equal info	Both
Dispersed support	81.56%	83.76%	78.43%	77.02%
Blockholder support	87.21%	83.21%	87.94%	91.83%
Total support	83.50%	83.90%	81.57%	81.40%
Var(Total support)	2.04%	2.39%	1.36%	1.40%
Passing rate	85.80%	84.79%	85.50%	84.59%
Close rate	15.11%	12.29%	26.08%	23.06%

Panel B. By ISS recommendation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Full model		Equal prefs		Equal info		Both	
	ISS=0	ISS=1	ISS=0	ISS=1	ISS=0	ISS=1	ISS=0	ISS=1
Dispersed support	61.82%	86.11%	65.11%	88.06%	57.43%	83.27%	55.56%	81.97%
Blockholder support	57.10%	94.14%	37.01%	93.85%	84.30%	88.77%	86.83%	92.98%
Total support	61.05%	88.68%	58.09%	89.85%	65.80%	85.21%	64.86%	85.21%
Var(Total support)	2.77%	0.44%	2.82%	0.40%	1.82%	0.55%	1.88%	0.51%
Passing rate	31.18%	98.39%	24.19%	98.76%	38.71%	96.28%	32.80%	96.53%
Close rate	41.40%	9.05%	33.87%	7.31%	55.38%	19.33%	51.61%	16.48%

A. Model Appendix

A.1. Equilibrium Posterior Densities

Lemma A.1 (Dispersed shareholder posterior density). *Symmetry in strategies, information, and preferences implies that each dispersed shareholder's best-response condition is the same, so there is a single limit to derive. As $N_D \to \infty$, the posterior density for any x_j converges as follows:*

$$\lim_{N_D \to \infty} f\big(x_j \mid z_{Dj},\, \eta_{jD},\, \mathrm{piv}_{Dj}\big) \propto$$

$$f(x_j \mid \eta_{jD}) \sum_{V_{Bj}} \underbrace{f\big(z_{Dj} \mid x_j\big)}_{\textit{signal effect}} \; \underbrace{\Pr\big(V_{Bj} \mid x_j\big)}_{\textit{strategic effect}} \; \underbrace{\mathrm{dirac}\big(p_D(x_j, \eta_{jD}) - \tau^{V_{Bj}}\big)}_{\textit{pivotal constraint}} \; = \tag{A.1a}$$

$$\sum_{V_{Bj}} \underbrace{f\big(x^{V_{Bj}}_{\eta_{jD}} \mid \eta_{jD}\big)}_{\textit{prior (given } \eta_{jD})} \; \underbrace{f\big(z_{Dj} \mid x^{V_{Bj}}_{\eta_{jD}}\big)}_{\textit{signal effect}} \; \underbrace{\chi^{V_{Bj}}}_{\textit{Jacobian}} \; \underbrace{\Pr\big(V_{Bj} \mid x^{V_{Bj}}_{\eta_{jD}}\big)}_{\textit{strategic effect}} \; \mathrm{dirac}\big(x_j - x^{V_{Bj}}_{\eta_{jD}}\big)\,. \tag{A.1b}$$

The following objects are required:

- ***Dispersed voting probability:***

$$p_D(x_j, \eta_{jD}) = \Phi\left(\sigma_D\left[\frac{\eta_{jD}}{\sigma^2_{u_D}} + \frac{x_j}{\sigma^2_D} - k_D\Lambda_D\right]\right), \qquad \Lambda_D = 1 + \frac{1}{\sigma^2_{u_D}} + \frac{1}{\sigma^2_D},$$

  where σ^2_D is the variance of dispersed shareholders' private signal noise and $\sigma^2_{u_D}$ is the variance of their public signal noise.

- ***Value of x_j when a dispersed shareholder is pivotal, given profile*** V_{Bj} *(i.e., the solution to $p_D(x_j, \eta_{jD}) = \tau^{V_{Bj}}$):*

$$x^{V_{Bj}}_{\eta_{jD}} \;=\; \sigma^2_D\left(\Lambda_D k_D - \frac{\eta_{jD}}{\sigma^2_{u_D}} + \frac{q^{V_{Bj}}}{\sigma_D}\right), \qquad q^{V_{Bj}} \equiv \Phi^{-1}\big(\tau^{V_{Bj}}\big).$$

- ***Jacobian correction term*** $\mathrm{dirac}\big(p_D(x_j, \eta_{jD}) - \tau^{V_{Bj}}\big) \mapsto x_j = x^{V_{Bj}}_{\eta_{jD}}$*:*

$$\chi^{V_{Bj}} \;=\; \frac{\sigma_D}{\phi\big(q^{V_{Bj}}\big)}.$$

- ***Threshold dispersed share:***

$$\tau^{V_{Bj}} = \frac{\lambda^* - \Psi_{for}(V_{Bj})}{\Psi_D},$$

which is the dispersed support rate required for the proposal to pass given the blockholder vote profile V_{Bj}.

- ***Blockholder vote profile likelihood (integrating over unobserved η_{jB}):***

Blockholders observe a common public signal

$$\eta_{jB} = x_j + u_{jB}, \qquad u_{jB} \sim N(0, \sigma^2_{u_B}),$$

which dispersed shareholders do not observe. Conditional on (x_j, η_{jB}), the likelihood of a blockholder vote profile V_{Bj} is

$$\Pr\left(V_{Bj} \mid x_j, \eta_{jB}\right) = \prod_{b=1}^{N_B} \left[\Phi\left(A_b \eta_{jB} + B_b x_j - C_b\right)\right]^{V_{Bj}(b)} \left[1 - \Phi\left(A_b \eta_{jB} + B_b x_j - C_b\right)\right]^{1-V_{Bj}(b)},$$

where

$$A_b = \frac{\sigma_b}{\sigma^2_{u_B}}, \qquad B_b = \frac{1}{\sigma_b}, \qquad C_b = \sigma_b \Lambda_b k_b, \qquad \Lambda_b = 1 + \frac{1}{\sigma^2_{u_B}} + \frac{1}{\sigma^2_b},$$

and σ^2_b is blockholder b's private signal variance.

From the perspective of a dispersed shareholder (who does not observe η_{jB}), the relevant strategic weight is the profile likelihood integrated over *the unobserved blockholder public signal:*

$$\Pr\left(V_{Bj} \mid x_j\right) = \int \Pr\left(V_{Bj} \mid x_j, \eta_{jB}\right) f(\eta_{jB} \mid x_j)\, d\eta_{jB},$$

where $f(\eta_{jB} \mid x_j)$ is the Gaussian density implied by $\eta_{jB} = x_j + u_{jB}$ and $u_{jB} \sim N(0, \sigma^2_{u_B})$.

Proof. See Appendix A.2. ■

Remark on Lemma A.1. The Dirac delta function in (A.1a) enforces the pivotality condition for dispersed shareholders by restricting the posterior density to values of the latent proposal quality x_j that make a dispersed voter exactly pivotal: those for which the implied dispersed support rate $p_D(x_j, \eta_{jD})$ equals the threshold share $\tau^{V_{Bj}}$ associated with a particular blockholder vote profile V_{Bj}. Intuitively, as $N_D \to \infty$, there is a unique draw of x_j (conditional on the dispersed public signal η_{jD}) for which a dispersed shareholder is pivotal, and the posterior density tilts strongly towards these knife-edge states.

Each such state corresponds to one blockholder vote profile and its associated quality $x_{\eta_{jD}}^{V_{Bj}}$. Its contribution is weighted by the (conditional) prior density $f(x_{\eta_{jD}}^{V_{Bj}} \mid \eta_{jD})$, the likelihood of the dispersed private signal $f(z_{Dj} \mid x_{\eta_{jD}}^{V_{Bj}})$, the Jacobian term $\chi^{V_{Bj}}$, and the strategic likelihood $\Pr\left(V_{Bj} \mid x_{\eta_{jD}}^{V_{Bj}}\right)$, which already averages over the unobserved blockholder public signal η_{jB}.

Lemma A.2 (Blockholder posterior density)**.** *Given differences in preferences and information, each blockholder b may have a separate posterior density over the state. As $N_D \to \infty$, the conditional b-posterior density for any x_j converges to*

$$\lim_{N_D\to\infty} f\left(x_j \mid z_{bj}, \eta_{jB}, \mathrm{piv}_{bj}\right) \propto f(x_j \mid \eta_{jB}) \underbrace{f\left(z_{bj} \mid x_j\right)}_{\text{signal effect}} \sum_{V_{B,-b,j}} \underbrace{\Pi^{V_{B,-b,j}}(x_j)}_{\text{pivotality}} \underbrace{\Pr\left(V_{B,-b,j} \mid x_j, \eta_{jB}\right)}_{\text{strategic effect}}. \tag{A.2}$$

Here $f(x_j \mid \eta_{jB})$ is the normal posterior implied by the prior $x_j \sim N(0,1)$ and the blockholder public signal η_{jB}.

The objects appearing in (A.2) are:

- ***Pivotal-bounding dispersed support rates given*** $V_{B,-b,j}$***:***

$$\tau_L^{V_{B,-b,j}} = \frac{\lambda^* - \Psi_{for}(V_{B,-b,j}) - \psi_b}{\Psi_D}, \qquad \tau_H^{V_{B,-b,j}} = \frac{\lambda^* - \Psi_{for}(V_{B,-b,j})}{\Psi_D}.$$

  Blockholder b is pivotal when the dispersed support rate satisfies $\tau_j \in [\tau_L^{V_{B,-b,j}}, \tau_H^{V_{B,-b,j}}]$.

- ***Dispersed voting probability:*** *for any (x_j, η_{jD}), the limiting dispersed support rate is*

$$p_D(x_j, \eta_{jD}) = \Phi\left(\sigma_D\left[\frac{\eta_{jD}}{\sigma_{u_D}^2} + \frac{x_j}{\sigma_D^2} - k_D\Lambda_D\right]\right), \qquad \Lambda_D = 1 + \frac{1}{\sigma_{u_D}^2} + \frac{1}{\sigma_D^2},$$

  where σ_D^2 is the dispersed private signal variance.

- ***Pivotality probability for blockholder*** b***:*** *in the large-N_D limit, the dispersed support satisfies $\tau_j = p_D(x_j, \eta_{jD})$ almost surely. For a given blockholder profile $V_{B,-b,j}$, the probability that*

b is pivotal depends on x_j only through

$$\Pi^{V_{B,-b,j}}(x_j) = \Pr\left(\tau_j \in [\tau_L^{V_{B,-b,j}},\ \tau_H^{V_{B,-b,j}}] \,\middle|\, x_j \right).$$

Because $p_D(x_j, \eta_{jD})$ is strictly increasing in η_{jD} for fixed x_j, this condition is equivalent to η_{jD} lying in an interval $[\eta_L^{V_{B,-b,j}}(x_j),\ \eta_H^{V_{B,-b,j}}(x_j)]$, where

$$\eta_L^{V_{B,-b,j}}(x_j) = \sigma_{u_D}^2 \left(\Lambda_D k_D - \frac{x_j}{\sigma_D^2} + \frac{q_L^{V_{B,-b,j}}}{\sigma_D} \right), \qquad \eta_H^{V_{B,-b,j}}(x_j) = \sigma_{u_D}^2 \left(\Lambda_D k_D - \frac{x_j}{\sigma_D^2} + \frac{q_H^{V_{B,-b,j}}}{\sigma_D} \right),$$

with $q_L^{V_{B,-b,j}} = \Phi^{-1}\left(\tau_L^{V_{B,-b,j}}\right)$ and $q_H^{V_{B,-b,j}} = \Phi^{-1}\left(\tau_H^{V_{B,-b,j}}\right)$. Let $f(\eta_{jD} \mid x_j)$ denote the conditional density of the dispersed public signal implied by $\eta_{jD} = x_j + u_{jD}$. Then

$$\Pi^{V_{B,-b,j}}(x_j) = \int_{\eta_L^{V_{B,-b,j}}(x_j)}^{\eta_H^{V_{B,-b,j}}(x_j)} f(\eta_{jD} \mid x_j)\, d\eta_{jD}.$$

- ***Blockholder voting profile (excluding b) likelihood, conditional on (x_j, η_{jB}):** blockholders share the common public signal η_{jB} and have private signal variances $\{\sigma_{b'}^2\}_{b'=1}^{N_B}$. Conditional on (x_j, η_{jB}), the likelihood of observing a vote profile $V_{B,-b,j}$ from the other blockholders is*

$$\Pr\left(V_{B,-b,j} \mid x_j, \eta_{jB}\right) = \prod_{\substack{b'=1 \\ b' \neq b}}^{N_B} \left[\Phi\left(A_{b'}\eta_{jB} + B_{b'}x_j - C_{b'}\right)\right]^{V_{B,-b,j}(b')} \left[1 - \Phi\left(A_{b'}\eta_{jB} + B_{b'}x_j - C_{b'}\right)\right]^{1 - V_{B,-b,j}(b')},$$

where

$$A_{b'} = \frac{\sigma_{b'}}{\sigma_{u_B}^2}, \qquad B_{b'} = \frac{1}{\sigma_{b'}}, \qquad C_{b'} = \sigma_{b'}\Lambda_{b'}k_{b'}, \qquad \Lambda_{b'} = 1 + \frac{1}{\sigma_{u_B}^2} + \frac{1}{\sigma_{b'}^2}.$$

Proof. See Appendix A.2. ■

Remark on Lemma A.2. Given the voting profile of the other blockholders $\mathbf{V}_{B_j}^{-b}$, the fraction of shares cast in favor by all shareholders excluding b is $s_B^{-b} + \tau_j \Psi_D$. Blockholder b is pivotal when this lies in $[\lambda^* - \psi_b,\ \lambda^*]$, which is equivalent to the dispersed support rate falling in $[\tau_L^{V_{B,-b,j}},\ \tau_H^{V_{B,-b,j}}]$.

In the large-N_D limit, the dispersed support rate equals $\tau_j = p_D(x_j, \eta_{jD})$ for a given (x_j, η_{jD}).

However, the blockholder does not observe η_{jD}; instead, she knows its distribution conditional on x_j through the dispersed public signal process. Thus, from her perspective, pivotality for a given blockholder profile $V_{B,-b,j}$ occurs with probability $\Pi^{V_{B,-b,j}}(x_j)$, which integrates over the range of dispersed public signals that would make her pivotal.

The posterior in (A.2) therefore has three components: (i) a prior $f(x_j \mid \eta_{jB})$ shaped by the blockholder's own public signal, (ii) a private signal likelihood $f(z_{bj} \mid x_j)$, and (iii) a strategic term combining the likelihood of the other blockholders' votes given (x_j, η_{jB}) with the pivotality probability $\Pi^{V_{B,-b,j}}(x_j)$, which summarizes how dispersed public information affects the event that blockholder b is decisive.

A.2. Proofs of Lemmas A.1 and A.2

Proof of Lemma A.1. Fix a proposal j. By symmetry, all dispersed shareholders share the same information structure and strategy, so it suffices to derive the posterior for a representative dispersed shareholder.

1. **Bayes rule and separation of signal vs. pivotality.** Consider the posterior conditional on the dispersed shareholder's private signal z_{Dj}, the dispersed public signal η_{jD}, and the event of pivotality piv_{Dj}. By Bayes' rule,

$$f(x_j \mid z_{Dj}, \eta_{jD}, \text{piv}_{Dj}) \;\propto\; f(x_j \mid \eta_{jD})\, f(z_{Dj} \mid x_j)\, \Pr(\text{piv}_{Dj} \mid x_j, z_{Dj}, \eta_{jD}). \tag{A.3}$$

   Conditional on x_j, the dispersed private signal z_{Dj} is independent of η_{jD} and of other shareholders' votes, so $\Pr(\text{piv}_{Dj} \mid x_j, z_{Dj}, \eta_{jD}) = \Pr(\text{piv}_{Dj} \mid x_j, \eta_{jD})$. Substituting into (A.3) yields

$$f(x_j \mid z_{Dj}, \eta_{jD}, \text{piv}_{Dj}) \;\propto\; f(x_j \mid \eta_{jD})\, f(z_{Dj} \mid x_j)\, \Pr(\text{piv}_{Dj} \mid x_j, \eta_{jD}). \tag{A.4}$$

2. **Decomposition of pivotality across blockholder vote profiles.** Let $V_{Bj} \in \{0,1\}^{N_B}$ denote the blockholder vote profile. Conditioning on V_{Bj} and using the law of total probability,

$$\Pr(\text{piv}_{Dj} \mid x_j, \eta_{jD}) = \sum_{V_{Bj}} \Pr(\text{piv}_{Dj} \mid x_j, \eta_{jD}, V_{Bj})\, \Pr(V_{Bj} \mid x_j, \eta_{jD}). \tag{A.5}$$

From the dispersed shareholder's perspective, the blockholders' public signal η_{jB} is unobserved. Under the maintained information structure, conditional on x_j the dispersed signal η_{jD} carries no additional information about η_{jB} beyond x_j; hence $\Pr(V_{Bj} \mid x_j, \eta_{jD}) = \Pr(V_{Bj} \mid x_j)$, where

$$\Pr(V_{Bj} \mid x_j) = \int \Pr(V_{Bj} \mid x_j, \eta_{jB})\, f(\eta_{jB} \mid x_j)\, d\eta_{jB}.$$

(If your model allows correlation between η_{jD} and η_{jB} conditional on x_j, the same argument goes through with $f(\eta_{jB} \mid x_j, \eta_{jD})$ instead.)

3. **Large-N_D limit and the pivotality constraint.** Fix (x_j, η_{jD}). Given the symmetric dispersed strategy, dispersed votes are i.i.d. conditional on (x_j, η_{jD}) and satisfy

$$\frac{1}{N_D} \sum_{i=1}^{N_D} \mathbf{1}\{V_{ij} = 1\} \xrightarrow[N_D \to \infty]{a.s.} p_D(x_j, \eta_{jD}),$$

where $p_D(x_j, \eta_{jD})$ is the dispersed voting probability stated in the proposition. A dispersed shareholder is pivotal under profile V_{Bj} only when the dispersed support rate (excluding her own infinitesimal vote) equals the threshold share $\tau^{V_{Bj}}$. In the limit $N_D \to \infty$, this becomes the knife-edge condition

$$p_D(x_j, \eta_{jD}) = \tau^{V_{Bj}}.$$

Because this event has probability zero under a continuous signal structure, we represent the limiting contribution of pivotality using a Dirac delta. Thus,

$$\Pr(\mathrm{piv}_{Dj} \mid x_j, \eta_{jD}, V_{Bj}) \;\propto\; \mathrm{dirac}\big(p_D(x_j, \eta_{jD}) - \tau^{V_{Bj}}\big)\,.$$

Substituting into (A.4)–(A.5) yields

$$f(x_j \mid z_{Dj}, \eta_{jD}, \mathrm{piv}_{Dj}) \;\propto\; f(x_j \mid \eta_{jD}) \sum_{V_{Bj}} f(z_{Dj} \mid x_j)\, \Pr(V_{Bj} \mid x_j)\, \mathrm{dirac}\big(p_D(x_j, \eta_{jD}) - \tau^{V_{Bj}}\big)\,,$$

which is (A.1a).

4. **Collapse to an atomic posterior via the Dirac delta identity.** For fixed η_{jD} and V_{Bj},

the mapping $x_j \mapsto p_D(x_j, \eta_{jD})$ is strictly increasing, so the equation $p_D(x_j, \eta_{jD}) = \tau^{V_{Bj}}$ has a unique solution $x_{\eta_{jD}}^{V_{Bj}}$ given in the proposition. Let $q^{V_{Bj}} = \Phi^{-1}(\tau^{V_{Bj}})$. Then at the root,

$$q^{V_{Bj}} = \sigma_D \left[\frac{\eta_{jD}}{\sigma_{u_D}^2} + \frac{x_{\eta_{jD}}^{V_{Bj}}}{\sigma_D^2} - k_D \Lambda_D \right].$$

Moreover,

$$\frac{\partial}{\partial x_j} p_D(x_j, \eta_{jD}) = \phi\left(\sigma_D \left[\frac{\eta_{jD}}{\sigma_{u_D}^2} + \frac{x_j}{\sigma_D^2} - k_D \Lambda_D \right] \right) \cdot \sigma_D \cdot \frac{1}{\sigma_D^2} = \frac{1}{\sigma_D} \phi\left(\sigma_D \left[\frac{\eta_{jD}}{\sigma_{u_D}^2} + \frac{x_j}{\sigma_D^2} - k_D \Lambda_D \right] \right).$$

Evaluated at $x_j = x_{\eta_{jD}}^{V_{Bj}}$, this derivative equals $\phi(q^{V_{Bj}})/\sigma_D$. Using the standard identity

$$\mathrm{dirac}(g(x)) = \sum_{x^\star : g(x^\star)=0} \frac{\mathrm{dirac}(x - x^\star)}{|g'(x^\star)|},$$

with $g(x) = p_D(x, \eta_{jD}) - \tau^{V_{Bj}}$, we obtain

$$\mathrm{dirac}\big(p_D(x_j, \eta_{jD}) - \tau^{V_{Bj}}\big) = \frac{\sigma_D}{\phi(q^{V_{Bj}})} \mathrm{dirac}\big(x_j - x_{\eta_{jD}}^{V_{Bj}}\big) \equiv \chi^{V_{Bj}} \mathrm{dirac}\big(x_j - x_{\eta_{jD}}^{V_{Bj}}\big).$$

Substituting this into (A.1a) yields (A.1b). This completes the proof.

■

Proof of Lemma A.2. Fix a proposal j and a focal blockholder b. Unlike dispersed shareholders, blockholders may have heterogeneous preferences and private signal variances, so we derive the posterior for a given b.

1. **Bayes rule.** By Bayes' rule,

$$f(x_j \mid z_{bj}, \eta_{jB}, \mathrm{piv}_{bj}) \;\propto\; f(x_j \mid \eta_{jB})\, f(z_{bj} \mid x_j)\, \Pr(\mathrm{piv}_{bj} \mid x_j, z_{bj}, \eta_{jB}). \tag{A.6}$$

Conditional on x_j, the private signal z_{bj} is independent of the remaining shareholders' votes and of η_{jD}, so $\Pr(\mathrm{piv}_{bj} \mid x_j, z_{bj}, \eta_{jB}) = \Pr(\mathrm{piv}_{bj} \mid x_j, \eta_{jB})$. Hence

$$f(x_j \mid z_{bj}, \eta_{jB}, \mathrm{piv}_{bj}) \;\propto\; f(x_j \mid \eta_{jB})\, f(z_{bj} \mid x_j)\, \Pr(\mathrm{piv}_{bj} \mid x_j, \eta_{jB}). \tag{A.7}$$

2. **Decompose pivotality over other-blockholder vote profiles.** Let $V_{B,-b,j} \in \{0,1\}^{N_B-1}$ denote the vote profile of all blockholders except b. By the law of total probability,

$$\Pr(\text{piv}_{bj} \mid x_j, \eta_{jB}) = \sum_{V_{B,-b,j}} \Pr(\text{piv}_{bj} \mid x_j, \eta_{jB}, V_{B,-b,j}) \Pr(V_{B,-b,j} \mid x_j, \eta_{jB}). \qquad \text{(A.8)}$$

Conditional on (x_j, η_{jB}), the other blockholders' votes are independent under the cutoff (probit) strategies, which delivers the product form for $\Pr(V_{B,-b,j} \mid x_j, \eta_{jB})$ stated in the proposition.

3. **Characterize pivotality conditional on $V_{B,-b,j}$.** Fix $(x_j, \eta_{jB}, V_{B,-b,j})$. Blockholder b is pivotal when the total fraction of shares cast in favor by all shareholders other than b lies in the interval $[\lambda^* - \psi_b, \lambda^*]$. Equivalently, the dispersed support rate τ_j must satisfy

$$\tau_j \in [\tau_L^{V_{B,-b,j}},\ \tau_H^{V_{B,-b,j}}],$$

where $\tau_L^{V_{B,-b,j}}$ and $\tau_H^{V_{B,-b,j}}$ are defined in the proposition.

As $N_D \to \infty$, conditional on (x_j, η_{jD}) the dispersed support rate satisfies $\tau_j = p_D(x_j, \eta_{jD})$ almost surely by the law of large numbers. The focal blockholder does not observe η_{jD}, but she knows its conditional distribution given x_j implied by $\eta_{jD} = x_j + u_{jD}$. Therefore,

$$\begin{aligned}\Pr(\text{piv}_{bj} \mid x_j, \eta_{jB}, V_{B,-b,j}) &= \Pr\left(\tau_j \in [\tau_L^{V_{B,-b,j}}, \tau_H^{V_{B,-b,j}}] \,\middle|\, x_j, \eta_{jB}, V_{B,-b,j}\right) \\ &= \Pr\left(p_D(x_j, \eta_{jD}) \in [\tau_L^{V_{B,-b,j}}, \tau_H^{V_{B,-b,j}}] \,\middle|\, x_j\right) \equiv \Pi^{V_{B,-b,j}}(x_j), \qquad \text{(A.9)}\end{aligned}$$

where the second equality uses that η_{jD} is conditionally independent of η_{jB} given x_j under the maintained signal structure.

4. **Express $\Pi^{V_{B,-b,j}}(x_j)$ as an integral over η_{jD}.** For fixed x_j, $p_D(x_j, \eta_{jD})$ is strictly increasing in η_{jD}. Hence the event $p_D(x_j, \eta_{jD}) \in [\tau_L^{V_{B,-b,j}}, \tau_H^{V_{B,-b,j}}]$ is equivalent to $\eta_{jD} \in [\eta_L^{V_{B,-b,j}}(x_j),\ \eta_H^{V_{B,-b,j}}(x_j)]$, where $\eta_L^{V_{B,-b,j}}(x_j)$ and $\eta_H^{V_{B,-b,j}}(x_j)$ are obtained by inverting $p_D(x_j, \eta_{jD}) = \tau_L^{V_{B,-b,j}}$ and $p_D(x_j, \eta_{jD}) =$

$\tau_H^{V_{B,-b,j}}$, respectively, yielding the expressions in the proposition. Therefore,

$$\Pi^{V_{B,-b,j}}(x_j) = \int_{\eta_L^{V_{B,-b,j}}(x_j)}^{\eta_H^{V_{B,-b,j}}(x_j)} f(\eta_{jD} \mid x_j)\, d\eta_{jD}.$$

5. **Combine pieces to obtain the posterior.** Substituting (A.9) into (A.8) gives

$$\Pr(\mathrm{piv}_{bj} \mid x_j, \eta_{jB}) = \sum_{V_{B,-b,j}} \Pi^{V_{B,-b,j}}(x_j)\, \Pr(V_{B,-b,j} \mid x_j, \eta_{jB}).$$

   Plugging this into (A.7) yields

$$f(x_j \mid z_{bj}, \eta_{jB}, \mathrm{piv}_{bj}) \;\propto\; f(x_j \mid \eta_{jB})\, f(z_{bj} \mid x_j) \sum_{V_{B,-b,j}} \Pi^{V_{B,-b,j}}(x_j)\, \Pr(V_{B,-b,j} \mid x_j, \eta_{jB}),$$

   which is exactly (A.2). This completes the proof.

■

B. Estimation Appendix

B.1. Identification Strategy

This section formally outlines the identification strategy. We first begin with the formal proof of Proposition 1, which references the lemmas following.

Proof of Proposition 1. The proof proceeds by showing how each block of parameters is pinned down by a particular part of the joint distribution of the observables and then invoking the corresponding lemma.

- **Advisor alignment and threshold.** Using the joint distribution of (T_j, S_j), we consider the conditional recommendation probability $P(S_j = 1 \mid T_j = t)$. Under the Gaussian structure, this is a probit in t whose slope and intercept are one-to-one functions of the advisor–alignment parameter ρ and the scaled threshold $\xi/\sqrt{1-\rho^2}$. With $\mathrm{Var}(s_j)$ normalized, this identifies both ρ and the threshold ξ. See Lemma B.1.

- **Dispersed information and cutoff.** From the dispersed support rate τ_j we construct the continuous index $T_j = \Phi^{-1}(\tau_j)$. The joint distribution of (τ_j, S_j) (equivalently (T_j, S_j))

identifies the dispersed-side information parameters $(\sigma^2_{uD}, \sigma^2_{\varepsilon D})$ and the dispersed cutoff k_D by exploiting: (i) the Gaussian index structure of T_j in the latent signals; and (ii) the way the distribution of T_j is truncated and shifted across $S_j = 0$ versus $S_j = 1$. Formally, see Lemma B.2.

- **State distribution given (τ_j, S_j).** Given the identified dispersed information parameters and advisor primitives $(\sigma^2_{uD}, \sigma^2_{\varepsilon D}, k_D, \rho, \xi)$, the Gaussian structure implies that the latent state conditional on (τ_j, S_j) is normal, with mean and variance that can be written explicitly as functions of these parameters. Intuitively, (τ_j, S_j) summarize how the underlying state loads into dispersed voting and the advisor's assessment. The resulting expressions $\mu_{x|\tau,\ell}(\tau_j)$ and $\sigma^2_{x|\tau,\ell}(\tau_j)$ for $\ell \in \{0,1\}$ are given in Lemma B.3.

- **Blockholder public information.** Conditional on (τ_j, S_j), the state x_j has the normal distribution characterized above. Blockholders observe a group-specific public signal $\eta_{Bj} = x_j + u_{Bj}$ with variance σ^2_{uB}. The way aggregate blockholder support $s_{Bj} = \sum_b \psi_b V_{bj}$ moves with (T_j, S_j) and how dispersed blockholder votes are around that aggregate identify the reduced-form blockholder public variance σ^2_{uB}: more precise public information (smaller σ^2_{uB}) makes s_{Bj} track (T_j, S_j) more tightly, while less precise information makes blockholder support more diffuse even conditional on (τ_j, S_j). This argument is formalized in Lemma B.4.

- **Blockholder private information and cutoffs.** Once σ^2_{uB} is known, each blockholder b's vote can be written, after integrating out unobserved signals, as

$$\Pr(V_{bj} = 1 \mid T_j, S_j) = \Phi\big(\alpha_{b0} + \alpha_{b1} T_j + \alpha_{b2} S_j\big),$$

for some coefficients $(\alpha_{b0}, \alpha_{b1}, \alpha_{b2})$ that are smooth functions of $(\sigma^2_{uB}, \sigma^2_{\varepsilon_b}, k_b)$ and the already-identified dispersed parameters. The variation in (T_j, S_j) across proposals identifies $(\alpha_{b0}, \alpha_{b1}, \alpha_{b2})$ from the observed probit of V_{bj} on (T_j, S_j), and the structural mapping from $(\alpha_{b0}, \alpha_{b1}, \alpha_{b2})$ to $(\sigma^2_{\varepsilon_b}, k_b)$ is one-to-one. Thus each blockholder's private-signal variance and cutoff are identified. See Lemma B.4.

- **Preferences.** Finally, given the identified information environment and cutoffs

$$(\sigma^2_{uD}, \sigma^2_{\varepsilon D}, k_D,\ \sigma^2_{uB}, \{\sigma^2_{\varepsilon_b}, k_b\}_b,\ \rho, \xi),$$

  the equilibrium pivotality probabilities and the posterior distribution of the state at each cutoff are fully determined by the model. For the dispersed shareholders,

$$\delta_D = -\,\mathrm{E}\big[x_j \mid m_{Dj} = k_D,\ \mathrm{piv}_{Dj}\big],$$

  where the expectation on the right-hand side is a known functional of the identified primitives (obtained by combining the Gaussian law of m_{Dj} with the equilibrium pivotality kernel). An analogous expression holds for each blockholder:

$$\delta_b = -\,\mathrm{E}\big[x_j \mid m_{bj} = k_b,\ \mathrm{piv}_{bj}\big].$$

  Lemma B.5 shows that these expectations are uniquely determined by the previously identified parameters, so δ_D and $\{\delta_b\}_b$ are identified.

Collecting these steps, we obtain that the mapping from the joint distribution of the observables $\{S_j, \tau_j, \{V_{bj}, \psi_b\}_b\}_j$ to the structural parameter vector is one-to-one, which establishes Proposition 1.

■

Lemma B.1 (Identification of advisor alignment ρ and threshold ξ)**.** *Let $T_j = \Phi^{-1}(\tau_j)$ denote the dispersed voting index for proposal j and suppose that there exists a latent Gaussian advisor index $\tilde{s}_j = x_j + u_j$,. Given x_j is assumed standard normal and $u_j \sim N(0, \sigma_I^2)$, we can normalize $\tilde{s}_j$ using variance $1/\sqrt{1+\sigma_I^2}$ such that s_j is also standard normal. This gives a bivariate normal:*

$$\begin{pmatrix} T_j \\ s_j \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_T \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_T^2 & \rho\,\sigma_T \\ \rho\,\sigma_T & 1 \end{pmatrix}\right),$$

with unknown correlation parameter $\rho \in (-1, 1)$ and variance normalization $\mathrm{Var}(s_j) = 1$. *The* *binary recommendation is generated by*

$$S_j = \mathbf{1}\{s_j > \xi\},$$

for some threshold $\xi \in \mathbb{R}$. The econometrician observes the joint distribution of (T_j, S_j) across proposals and knows the marginal distribution of T_j (hence μ_T and σ_T^2). Then the parameter ρ and the scaled threshold $\xi/\sqrt{1-\rho^2}$ are identified from the joint distribution of (T_j, S_j).

Sketch of proof. Standardizing T_j using its known mean and variance, define

$$\tilde{T}_j \equiv \frac{T_j - \mu_T}{\sigma_T}.$$

Then $(\tilde{T}_j, s_j)$ is bivariate normal with zero means, unit variances, and correlation ρ:

$$\begin{pmatrix} \tilde{T}_j \\ s_j \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right), \qquad S_j = \mathbf{1}\{s_j > \xi\}.$$

Conditional on $\tilde{T}_j = t$, the advisor index satisfies

$$s_j \mid \tilde{T}_j = t \sim N\big(\rho t,\ 1-\rho^2\big),$$

so

$$\Pr(S_j = 1 \mid \tilde{T}_j = t) = \Pr(s_j > \xi \mid \tilde{T}_j = t) = \Phi\left(\frac{\rho t - \xi}{\sqrt{1-\rho^2}}\right).$$

Define the inverse probit transform of this conditional probability,

$$g(t) \ \equiv\ \Phi^{-1}\big(\Pr(S_j = 1 \mid \tilde{T}_j = t)\big).$$

Then

$$g(t) = \frac{\rho t - \xi}{\sqrt{1-\rho^2}} = b_0 + b_1 t,$$

with slope $b_1 = \rho/\sqrt{1-\rho^2}$ and intercept $b_0 = -\xi/\sqrt{1-\rho^2}$.

The joint distribution of $(\tilde{T}_j, S_j)$ identifies the function $t \mapsto \Pr(S_j = 1 \mid \tilde{T}_j = t)$ and hence identifies $g(t)$. Because $g(t)$ is affine, its slope b_1 and intercept b_0 are uniquely determined. The slope satisfies

$$b_1 = \frac{\rho}{\sqrt{1-\rho^2}},$$

which defines a strictly monotone mapping $\rho \mapsto b_1$ on $(-1, 1)$; thus ρ is uniquely recovered from b_1, i.e.,

$$\rho = \frac{b_1}{\sqrt{1+b_1^2}}.$$

Given ρ, the scaled threshold $\xi/\sqrt{1-\rho^2}$ is then recovered from the intercept $b_0 = -\xi/\sqrt{1-\rho^2}$, i.e.,

$$\xi = -b_0\left(\sqrt{1-\frac{b_1}{\sqrt{1+b_1^2}}}\right).$$

Hence ρ and $\xi/\sqrt{1-\rho^2}$ are identified from the joint distribution of (T_j, S_j). ■

Lemma B.2 (Identification of dispersed shareholder parameters)**.** *Suppose the primitives of the model in Section 2 hold: the latent state is $x_j \sim N(0,1)$, dispersed shareholders observe the reduced-form public signal*

$$\eta_{Dj} = x_j + u_{Dj}, \qquad u_{Dj} \sim N(0, \sigma_{u_D}^2),$$

and each dispersed shareholder i receives a private signal $z_{ij} = x_j + \varepsilon_{ij}$ with $\varepsilon_{ij} \sim N(0, \sigma_{\varepsilon_D}^2)$, independent of (x_j, u_{Dj}). Let $p_D(x_j, \eta_{Dj})$ denote the dispersed voting probability implied by the voting rule in Equation (12) specialized to the dispersed type $(\sigma_{u_D}^2, \sigma_{\varepsilon_D}^2, k_D)$, and define

$$\tau_j \;=\; p_D(x_j, \eta_{Dj})$$

as the dispersed support rate for proposal j in the $N_D \to \infty$ limit.

In addition, suppose there exists a proxy advisor that forms a continuous assessment

$$s_j = x_j + u_{Ij}, \qquad u_{Ij} \sim N(0, \sigma_I^2),$$

independent of u_{Dj}, and issues the binary recommendation

$$S_j \;=\; \mathbb{1}\left[s_j > \xi\right],$$

where both the correlation between inverse probit $T_j = \Phi^{-1}(\tau_j)$ and the latent proxy advisor signal $\rho = \mathrm{Corr}\left(T_j, s_j\right)$ and ξ are identified from Lemma B.1.

The econometrician observes the joint cross-sectional distribution of (τ_j, S_j) across proposals. Then the dispersed public signal variance $\sigma^2_{u_D}$, the dispersed private signal variance $\sigma^2_{\varepsilon_D}$, and the dispersed cutoff k_D are identified.

Proof. Specializing Equation (12) to the dispersed type yields, for each proposal j,

$$\tau_j = \Pr\left(V_{Dj} = 1 \mid x_j, \eta_{Dj}\right) = \Phi\left(\sigma_{\varepsilon_D}\left[\frac{\eta_{Dj}}{\sigma^2_{u_D}} + \frac{x_j}{\sigma^2_{\varepsilon_D}} - k_D\Lambda_D\right]\right), \tag{B.1}$$

with $\sigma^2_{\varepsilon_D} = \sigma^2_{\varepsilon_D}$ and $\Lambda_D = 1 + 1/\sigma^2_{u_D} + 1/\sigma^2_{\varepsilon_D}$. Applying the inverse probit transform gives

$$T_j \;\equiv\; \Phi^{-1}(\tau_j) = \sigma_{\varepsilon_D}\left[\frac{\eta_{Dj}}{\sigma^2_{u_D}} + \frac{x_j}{\sigma^2_{\varepsilon_D}} - k_D\Lambda_D\right] \;=\; a\,\eta_{Dj} + b\,x_j + c, \tag{B.2}$$

where

$$a = \frac{\sigma_{\varepsilon_D}}{\sigma^2_{u_D}}, \qquad b = \frac{1}{\sigma_{\varepsilon_D}}, \qquad c = -\,\sigma_{\varepsilon_D} k_D \Lambda_D.$$

By assumption, $x_j \sim N(0,1)$, $\eta_{Dj} = x_j + u_{Dj}$ with $u_{Dj} \sim N(0, \sigma^2_{u_D})$, and $s_j = x_j + u_{Ij}$ with $u_{Ij} \sim N(0, \sigma^2_I)$, all mutually independent. Hence (x_j, η_{Dj}, s_j) is trivariate normal with

$$\mathrm{Var}(x_j) = 1, \quad \mathrm{Var}(\eta_{Dj}) = 1 + \sigma^2_{u_D}, \quad \mathrm{Var}(s_j) = 1 + \sigma^2_I,$$

and

$$\mathrm{Cov}(x_j, \eta_{Dj}) = 1, \quad \mathrm{Cov}(x_j, s_j) = 1, \quad \mathrm{Cov}(\eta_{Dj}, s_j) = 1.$$

Equation (B.2) shows that T_j is an affine function of (x_j, η_{Dj}), so the pair (T_j, s_j) is also jointly Gaussian. The recommendation rule $S_j = \mathbb{1}\left[s_j > \xi\right]$ implies that the econometrician observes (T_j, S_j), where S_j indicates whether the latent normal s_j lies above the threshold ξ.

Let $(\mu_T, \mu_s, \sigma^2_T, \sigma^2_s, \rho_{Ts})$ denote the mean, variances, and correlation of the bivariate normal

(T_j, s_j). The joint cross-sectional distribution of (T_j, S_j) identifies these five quantities. First, the marginal distribution of T_j is directly observed, so (μ_T, σ_T^2) are identified. Second, the marginal distribution of the binary S_j identifies the standardized cutoff of s_j:

$$\Pr(S_j = 1) = \Pr(s_j > \xi) = 1 - \Phi\left(\frac{\xi - \mu_s}{\sigma_s}\right),$$

so (μ_s, σ_s, ξ) are identified up to the normalization of s_j; with $\mathrm{Var}(x_j) = 1$ fixed, this pins down $\sigma_s^2 = 1 + \sigma_I^2$ and the standardized threshold ξ/σ_s. Third, the conditional means and variances of T_j given $S_j = 0$ and $S_j = 1$ are identified from the joint distribution of (T_j, S_j). For a bivariate normal (T, s) truncated along the s-dimension, these conditional moments are smooth functions of $(\mu_T, \mu_s, \sigma_T^2, \sigma_s^2, \rho_{Ts})$ and the threshold ξ; inverting these relationships yields ρ_{Ts} and hence the covariance $\mathrm{Cov}(T_j, s_j)$ in addition to μ_s and σ_s^2.

We now relate $(\sigma_T^2, \mathrm{Cov}(T_j, s_j))$ to the underlying structural parameters. From (B.2) and the covariance structure of (x_j, η_{Dj}, s_j), we obtain

$$\mathrm{Var}(T_j) = a^2\,\mathrm{Var}(\eta_{Dj}) + b^2\,\mathrm{Var}(x_j) + 2ab\;\mathrm{Cov}(\eta_{Dj}, x_j) = a^2(1 + \sigma_{u_D}^2) + b^2 + 2ab,$$

and

$$\mathrm{Cov}(T_j, s_j) = a\;\mathrm{Cov}(\eta_{Dj}, s_j) + b\;\mathrm{Cov}(x_j, s_j) = a \cdot 1 + b \cdot 1 = a + b,$$

where the last equality uses $\mathrm{Cov}(\eta_{Dj}, s_j) = \mathrm{Cov}(x_j, s_j) = 1$ under the assumed independence structure.[22]

The left-hand sides $(\sigma_T^2, \mathrm{Cov}(T_j, s_j))$ are identified from the joint distribution of (T_j, S_j), and the right-hand sides depend only on $(a, b, \sigma_{u_D}^2)$, with $a = \sigma_{\varepsilon_D}/\sigma_{u_D}^2$ and $b = 1/\sigma_{\varepsilon_D}$. This gives a system of two equations in the two unknowns $(\sigma_{u_D}^2, \sigma_{\varepsilon_D}^2)$; under the usual positivity restrictions on variances, the mapping

$$(\sigma_{u_D}^2, \sigma_{\varepsilon_D}^2) \mapsto (\sigma_T^2, \mathrm{Cov}(T_j, s_j))$$

[22] If instead η_{Dj} is written as a noisy transformation of s_j as in the information section, the same logic applies with $\mathrm{Cov}(\eta_{Dj}, s_j)$ replaced by $\mathrm{Var}(s_j)$; the reduced-form variance $\sigma_{u_D}^2$ then absorbs the advisor and group-specific components.

is one-to-one, so $(\sigma^2_{u_D}, \sigma^2_{\varepsilon_D})$ are identified.

Finally, the intercept c in (B.2) is the mean of T_j net of its dependence on (x_j, η_{Dj}):

$$c = \mathrm{E}[T_j] - a\,\mathrm{E}[\eta_{Dj}] - b\,\mathrm{E}[x_j] = \mu_T,$$

since $\mathrm{E}[x_j] = \mathrm{E}[\eta_{Dj}] = 0$ under our normalization. Given $(\sigma^2_{u_D}, \sigma^2_{\varepsilon_D})$ and $\Lambda_D = 1+1/\sigma^2_{u_D}+1/\sigma^2_{\varepsilon_D}$, we have

$$c = -\,\sigma_{\varepsilon_D} k_D \Lambda_D \quad \Longrightarrow \quad k_D = -\,\frac{c}{\sigma_{\varepsilon_D}\Lambda_D},$$

so k_D is identified. This completes the identification of $(\sigma^2_{u_D}, \sigma^2_{\varepsilon_D}, k_D)$ from the joint cross-sectional distribution of (τ_j, S_j). ■

Lemma B.3 (Conditional state distribution given (τ_j, S_j))**.** *Suppose the information structure in Section 2 holds. In particular, for each proposal j, the latent state satisfies $x_j \sim N(0,1)$, dispersed shareholders observe*

$$\eta_{Dj} = x_j + u_{Dj}, \qquad u_{Dj} \sim N(0, \sigma^2_{uD}),$$

and their private signals have variance $\sigma^2_{\varepsilon D}$. Let k_D denote the dispersed cutoff and define

$$T_j \equiv \Phi^{-1}(\tau_j), \qquad \Lambda_D \equiv 1 + \frac{1}{\sigma^2_{uD}} + \frac{1}{\sigma^2_{\varepsilon D}},$$

so that, from the dispersed voting rule,

$$T_j = \sigma_{\varepsilon D}\left[\frac{\eta_{Dj}}{\sigma^2_{uD}} + \frac{x_j}{\sigma^2_{\varepsilon D}} - k_D\Lambda_D\right]. \tag{B.3}$$

The proxy advisor observes a latent Gaussian index s_j and issues the recommendation

$$S_j = \mathbf{1}\{s_j > \xi\}.$$

We normalize $\mathrm{Var}(s_j) = 1$ *and let*

$$\rho \equiv \mathrm{Corr}(T_j, s_j)$$

denote the advisor–alignment parameter, which is identified from the joint distribution of (T_j, S_j)

as in Lemma B.1.

Then, for any proposal with interior $\tau_j \in (0,1)$ and for each $\ell \in \{0,1\}$, the conditional distribution of the state given the dispersed support rate and the recommendation is Gaussian:

$$x_j \mid (\tau_j, S_j = \ell) \sim N\left(\mu_{x|\tau,\ell}(\tau_j),\ \sigma^2_{x|\tau,\ell}(\tau_j)\right),$$

where the conditional means are

$$\mu_{x|\tau,1}(\tau_j) = \mu_{x|T}(t) + \frac{\sigma_{xs|T}}{\sigma_{s|T}}\,\lambda_1\big(\alpha(t)\big), \tag{B.4}$$

$$\mu_{x|\tau,0}(\tau_j) = \mu_{x|T}(t) - \frac{\sigma_{xs|T}}{\sigma_{s|T}}\,\lambda_0\big(\alpha(t)\big), \tag{B.5}$$

and the conditional variances are

$$\sigma^2_{x|\tau,1}(\tau_j) = \sigma^2_{x|T} - \frac{\sigma^2_{xs|T}}{\sigma^2_{s|T}}\,\delta_1\big(\alpha(t)\big), \tag{B.6}$$

$$\sigma^2_{x|\tau,0}(\tau_j) = \sigma^2_{x|T} - \frac{\sigma^2_{xs|T}}{\sigma^2_{s|T}}\,\delta_0\big(\alpha(t)\big), \tag{B.7}$$

with $t = \Phi^{-1}(\tau_j)$, and where

$$\begin{aligned}
\mu_T &\equiv \mathrm{E}[T_j] = -\,\sigma_{\varepsilon D} k_D \Lambda_D, \\
\sigma_{xT} &\equiv \mathrm{Cov}(x_j, T_j) = \sigma_{\varepsilon D}\left(\frac{1}{\sigma^2_{uD}} + \frac{1}{\sigma^2_{\varepsilon D}}\right), \\
\sigma^2_T &\equiv \mathrm{Var}(T_j) = \sigma^2_{\varepsilon D}\left[\left(\frac{1}{\sigma^2_{uD}} + \frac{1}{\sigma^2_{\varepsilon D}}\right)^2 + \frac{1}{\sigma^2_{uD}}\right].
\end{aligned}$$

The conditional mean and variance of x_j given $T_j = t$ are

$$\mu_{x|T}(t) = \mathrm{E}[x_j \mid T_j = t] = \frac{\sigma_{xT}}{\sigma^2_T}\,(t - \mu_T), \qquad \sigma^2_{x|T} = \mathrm{Var}(x_j \mid T_j) = 1 - \frac{\sigma^2_{xT}}{\sigma^2_T}.$$

Let $\sigma_{sT} \equiv \mathrm{Cov}(s_j, T_j) = \rho\,\sigma_T$, so that

$$\mu_{s|T}(t) = \mathrm{E}[s_j \mid T_j = t] = \frac{\sigma_{sT}}{\sigma^2_T}(t - \mu_T) = \frac{\rho}{\sigma_T}(t - \mu_T),$$

and

$$\sigma_{s|T}^2 = \mathrm{Var}(s_j \mid T_j) = 1 - \frac{\sigma_{sT}^2}{\sigma_T^2} = 1 - \rho^2, \qquad \sigma_{s|T} \equiv \sqrt{1-\rho^2}.$$

Under the structural representation $s_j = (x_j + u_{Ij})/\sqrt{1+\sigma_I^2}$ *with* u_{Ij} *independent of* (x_j, u_{Dj})*, we have* $\mathrm{Cov}(T_j, s_j) = \mathrm{Cov}(x_j, T_j)/\sqrt{1+\sigma_I^2}$ *and* $\mathrm{Cov}(x_j, s_j) = 1/\sqrt{1+\sigma_I^2}$*, which implies*

$$\sigma_{xs} \equiv \mathrm{Cov}(x_j, s_j) = \frac{\mathrm{Cov}(T_j, s_j)}{\mathrm{Cov}(x_j, T_j)} = \frac{\rho\,\sigma_T}{\sigma_{xT}}.$$

Hence the conditional covariance between x_j *and* s_j *given* T_j *is*

$$\begin{aligned}\sigma_{xs|T} \equiv \mathrm{Cov}(x_j, s_j \mid T_j) &= \sigma_{xs} - \frac{\sigma_{xT}\,\sigma_{sT}}{\sigma_T^2} \\ &= \frac{\rho\,\sigma_T}{\sigma_{xT}} - \frac{\sigma_{xT}\,\rho\,\sigma_T}{\sigma_T^2} = \rho\,\sigma_T\,\frac{\sigma_T^2 - \sigma_{xT}^2}{\sigma_{xT}\sigma_T^2}.\end{aligned}$$

The truncation index and inverse Mills ratios are

$$\alpha(t) = \frac{\xi - \mu_{s|T}(t)}{\sigma_{s|T}} = \frac{\xi - \frac{\rho}{\sigma_T}(t-\mu_T)}{\sqrt{1-\rho^2}},$$

$$\lambda_1(\alpha) = \frac{\phi(\alpha)}{1-\Phi(\alpha)}, \qquad \lambda_0(\alpha) = \frac{\phi(\alpha)}{\Phi(\alpha)},$$
$$\delta_1(\alpha) = \lambda_1(\alpha)\big(\lambda_1(\alpha) - \alpha\big), \qquad \delta_0(\alpha) = \lambda_0(\alpha)\big(\lambda_0(\alpha) + \alpha\big),$$

where $\phi(\cdot)$ *and* $\Phi(\cdot)$ *denote the standard normal PDF and CDF, respectively. All terms in* (B.4)–(B.7) *are thus explicit functions of the identified dispersed-side parameters* $(\sigma_{uD}^2, \sigma_{\varepsilon D}^2, k_D)$*, the advisor-alignment parameter* ρ*, and the threshold* ξ*.*

Proof. The linear representation (B.3) implies

$$T_j = a_x x_j + a_\eta \eta_{Dj} + c_D, \qquad a_x = \frac{1}{\sigma_{\varepsilon D}}, \quad a_\eta = \frac{\sigma_{\varepsilon D}}{\sigma_{uD}^2}, \quad c_D = -\,\sigma_{\varepsilon D} k_D \Lambda_D.$$

Since $x_j \sim N(0,1)$ and $\eta_{Dj} = x_j + u_{Dj}$ with $u_{Dj} \sim N(0, \sigma_{uD}^2)$ independent of x_j, straightforward calculations yield

$$\mu_T = \mathrm{E}[T_j] = c_D, \qquad \sigma_{xT} = \mathrm{Cov}(x_j, T_j) = a_x + a_\eta,$$

and

$$\sigma_T^2 = \mathrm{Var}(T_j) = a_x^2\,\mathrm{Var}(x_j) + a_\eta^2\,\mathrm{Var}(\eta_{Dj}) + 2a_x a_\eta\,\mathrm{Cov}(x_j, \eta_{Dj}) = \sigma_{\varepsilon D}^2\left[\left(\frac{1}{\sigma_{uD}^2} + \frac{1}{\sigma_{\varepsilon D}^2}\right)^2 + \frac{1}{\sigma_{uD}^2}\right].$$

Thus

$$\mu_{x|T}(t) = \mathrm{E}[x_j \mid T_j = t] = \frac{\sigma_{xT}}{\sigma_T^2}(t - \mu_T), \qquad \sigma_{x|T}^2 = 1 - \frac{\sigma_{xT}^2}{\sigma_T^2}$$

follow from standard Gaussian conditioning.

For the advisor, start from the structural latent index

$$\tilde{s}_j = x_j + u_{Ij}, \qquad u_{Ij} \sim N(0, \sigma_I^2),$$

independent of (x_j, u_{Dj}), and normalize $s_j = \tilde{s}_j/\sqrt{1+\sigma_I^2}$ so that $\mathrm{Var}(s_j) = 1$. Then

$$\mathrm{Cov}(x_j, \tilde{s}_j) = 1, \qquad \mathrm{Cov}(T_j, \tilde{s}_j) = \mathrm{Cov}(T_j, x_j),$$

because u_{Ij} is independent of (x_j, u_{Dj}) and T_j is linear in (x_j, u_{Dj}). Hence

$$\mathrm{Cov}(x_j, s_j) = \frac{1}{\sqrt{1+\sigma_I^2}}, \qquad \mathrm{Cov}(T_j, s_j) = \frac{\sigma_{xT}}{\sqrt{1+\sigma_I^2}}.$$

Letting $\rho = \mathrm{Corr}(T_j, s_j)$ and normalizing $\mathrm{Var}(s_j) = 1$, we have $\mathrm{Cov}(T_j, s_j) = \rho\,\sigma_T$, so

$$\frac{\sigma_{xT}}{\sqrt{1+\sigma_I^2}} = \mathrm{Cov}(T_j, s_j) = \rho\,\sigma_T \quad\Longrightarrow\quad \sqrt{1+\sigma_I^2} = \frac{\sigma_{xT}}{\rho\,\sigma_T}.$$

Substituting into $\mathrm{Cov}(x_j, s_j) = 1/\sqrt{1+\sigma_I^2}$ gives

$$\sigma_{xs} \equiv \mathrm{Cov}(x_j, s_j) = \frac{\rho\,\sigma_T}{\sigma_{xT}}.$$

Thus the joint vector (x_j, s_j, T_j) is trivariate normal with mean $(0, 0, \mu_T)'$ and covariance determined by $(\sigma_{xT}, \sigma_T^2, \rho)$ and the normalization $\mathrm{Var}(s_j) = 1$.

By the standard formula for conditioning in the multivariate normal, the conditional distribution of (x_j, s_j) given $T_j = t$ is bivariate normal with mean

$$\begin{pmatrix} \mu_{x|T}(t) \\ \mu_{s|T}(t) \end{pmatrix} = \begin{pmatrix} \sigma_{xT} \\ \sigma_{sT} \end{pmatrix} \frac{1}{\sigma_T^2}(t - \mu_T) = \begin{pmatrix} \frac{\sigma_{xT}}{\sigma_T^2}(t - \mu_T) \\ \frac{\rho}{\sigma_T}(t - \mu_T) \end{pmatrix},$$

and covariance matrix

$$\Sigma_{(x,s)|T} = \begin{pmatrix} 1 & \sigma_{xs} \\ \sigma_{xs} & 1 \end{pmatrix} - \begin{pmatrix} \sigma_{xT} \\ \sigma_{sT} \end{pmatrix} \frac{1}{\sigma_T^2} \begin{pmatrix} \sigma_{xT} & \sigma_{sT} \end{pmatrix}.$$

From this,

$$\sigma_{x|T}^2 = 1 - \frac{\sigma_{xT}^2}{\sigma_T^2}, \qquad \sigma_{s|T}^2 = 1 - \frac{\sigma_{sT}^2}{\sigma_T^2} = 1 - \rho^2,$$

and

$$\sigma_{xs|T} = \mathrm{Cov}(x_j, s_j \mid T_j) = \sigma_{xs} - \frac{\sigma_{xT}\sigma_{sT}}{\sigma_T^2} = \frac{\rho\,\sigma_T}{\sigma_{xT}} - \frac{\sigma_{xT}\,\rho\,\sigma_T}{\sigma_T^2} = \rho\,\sigma_T\,\frac{\sigma_T^2 - \sigma_{xT}^2}{\sigma_{xT}\sigma_T^2}.$$

Finally, conditioning on (τ_j, S_j) is equivalent to conditioning on $(T_j = t, S_j = \ell)$ with $t = \Phi^{-1}(\tau_j)$ and truncating on the s_j-dimension at $s_j > \xi$ (if $\ell = 1$) or $s_j \le \xi$ (if $\ell = 0$). Since $(x_j, s_j) \mid T_j = t$ is bivariate normal, the truncated normal formulas (see, e.g., Tallis, 1961) imply that

$$\mathrm{E}[x_j \mid T_j = t, S_j = 1] = \mu_{x|T}(t) + \frac{\sigma_{xs|T}}{\sigma_{s|T}}\,\lambda_1\big(\alpha(t)\big),$$

$$\mathrm{Var}(x_j \mid T_j = t, S_j = 1) = \sigma_{x|T}^2 - \frac{\sigma_{xs|T}^2}{\sigma_{s|T}^2}\,\delta_1\big(\alpha(t)\big),$$

and

$$\mathrm{E}[x_j \mid T_j = t, S_j = 0] = \mu_{x|T}(t) - \frac{\sigma_{xs|T}}{\sigma_{s|T}}\,\lambda_0\big(\alpha(t)\big),$$

$$\mathrm{Var}(x_j \mid T_j = t, S_j = 0) = \sigma_{x|T}^2 - \frac{\sigma_{xs|T}^2}{\sigma_{s|T}^2}\,\delta_0\big(\alpha(t)\big),$$

with

$$\alpha(t) = \frac{\xi - \mu_{s|T}(t)}{\sigma_{s|T}}, \quad \lambda_1(\alpha) = \frac{\phi(\alpha)}{1 - \Phi(\alpha)}, \quad \lambda_0(\alpha) = \frac{\phi(\alpha)}{\Phi(\alpha)},$$

and

$$\delta_1(\alpha) = \lambda_1(\alpha)\big(\lambda_1(\alpha) - \alpha\big), \qquad \delta_0(\alpha) = \lambda_0(\alpha)\big(\lambda_0(\alpha) + \alpha\big).$$

Replacing t with $\Phi^{-1}(\tau_j)$ yields (B.4)–(B.7) and completes the proof. ∎

Lemma B.4 (Identification of blockholder parameters). *Suppose the primitives of the model in Section 2 hold and that the dispersed side of the information structure has already been identified as in Lemma B.2. In particular:*

- *the state satisfies $x_j \sim N(0, 1)$;*
- *dispersed shareholders observe a reduced-form public signal*

$$\eta_{Dj} = x_j + u_{Dj}, \qquad u_{Dj} \sim N(0, \sigma^2_{u_D}),$$

  and private signals $z_{ij} = x_j + \varepsilon_{ij}$ with $\varepsilon_{ij} \sim N(0, \sigma^2_{\varepsilon_D})$;
- *a proxy advisor forms a latent Gaussian assessment s_j of x_j and issues the binary recommendation*

$$S_j = \mathbf{1}\{s_j > \xi\};$$

- *from the joint distribution of (τ_j, S_j) the econometrician has already identified $(\xi, \sigma^2_{u_D}, \sigma^2_{\varepsilon_D}, k_D)$, and hence knows the posterior law of x_j conditional on (τ_j, S_j):*

$$x_j \mid (\tau_j, S_j) \sim N\big(\mu_{x|\tau,S}(\tau_j, S_j), \sigma^2_{x|\tau,S}(\tau_j, S_j)\big),$$

  as implied by the Gaussian structure and the mapping between (x_j, η_{Dj}) and (τ_j, S_j) characterized in Lemma B.2.

On the blockholder side, suppose that for each blockholder $b = 1, \ldots, N_B$:

- *the blockholder-specific public signal is*

$$\eta_{Bj} = x_j + u_{Bj}, \qquad u_{Bj} \sim N(0, \sigma^2_{uB}),$$

independent of (x_j, u_{Dj})*;*

- *the private signal is*

$$z_{bj} = x_j + \varepsilon_{bj}, \qquad \varepsilon_{bj} \sim N(0, \sigma^2_{\varepsilon_b}),$$

independent across b *and of* (x_j, u_{Dj}, u_{Bj})*;*

- *blockholder* b *uses the cutoff strategy induced by the voting rule* (12)*:*

$$V_{bj} = 1 \quad \Longleftrightarrow \quad m_{bj} \equiv \mathbb{E}[x_j \mid \eta_{Bj}, z_{bj}] \geq k_b,$$

with type $(\sigma^2_{uB}, \sigma^2_{\varepsilon_b}, k_b)$*.*

Let $s_{Bj} = \sum_{b=1}^{N_B} \psi_b V_{bj}$ *denote aggregate blockholder support. Assume the econometrician observes, for each proposal j, the triple*

$$\big(\tau_j, S_j, s_{Bj}\big),$$

and, in addition, each individual blockholder vote V_{bj} *and ownership weight* ψ_b*.*

Then, under the Gaussian structure and the monotone index form of the voting rule (12)*, the joint cross-sectional distribution of*

$$\Big\{ \big(\tau_j, S_j, s_{Bj}, \{V_{bj}\}_{b=1}^{N_B}\big) \Big\}_j$$

identifies:

(i) the blockholder public-signal variance σ^2_{uB}*; and*

(ii) for each blockholder b*, the private-signal variance* $\sigma^2_{\varepsilon_b}$ *and cutoff* k_b*.*

Given these objects and the identified information structure on the dispersed side, each blockholder's preference parameter δ_b is then identified from the best-response condition

$$\mathbb{E}\left[x_j \,\middle|\, z_{bj} = k_b,\ \mathrm{piv}_{bj}\right] = -\delta_b$$

in (14).

Sketch of the proof. We prove the lemma with the following steps:

1. **Posterior of x_j and η_{Bj} given (τ_j, S_j).** By Lemma B.2, the parameters $(\xi, \sigma^2_{u_D}, \sigma^2_{\varepsilon_D}, k_D)$ are identified from the joint distribution of (τ_j, S_j), and the Gaussian structure implies a closed-form posterior

$$x_j \mid (\tau_j, S_j) \sim N\big(\mu_{x|\tau,S}(\tau_j, S_j), \sigma^2_{x|\tau,S}(\tau_j, S_j)\big).$$

   Given $\eta_{Bj} = x_j + u_{Bj}$ with $u_{Bj} \sim N(0, \sigma^2_{uB})$ independent of x_j, we obtain

$$\eta_{Bj} \mid (\tau_j, S_j) \sim N\Big(\mu_{x|\tau,S}(\tau_j, S_j),\ \sigma^2_{x|\tau,S}(\tau_j, S_j) + \sigma^2_{uB}\Big), \tag{B.8}$$

   which is the blockholder analogue of the conditional law of the dispersed public signal, with σ^2_{uB} entering only through the variance term.

2. **Identification of σ^2_{uB} from $s_{Bj} \mid (\tau_j, S_j)$.** Using (12), blockholder b's voting probability conditional on (x_j, η_{Bj}) is

$$p_b(x_j, \eta_{Bj}) = \Pr(V_{bj} = 1 \mid x_j, \eta_{Bj}) = \Phi\left(\sigma_{\varepsilon_b}\left[\frac{\eta_{Bj}}{\sigma^2_{uB}} + \frac{x_j}{\sigma^2_{\varepsilon_b}} - k_b\Lambda_b\right]\right),$$

   with $\Lambda_b = 1 + 1/\sigma^2_{uB} + 1/\sigma^2_{\varepsilon_b}$. Aggregate blockholder support is

$$s_{Bj} = \sum_{b=1}^{N_B} \psi_b V_{bj},$$

   so conditional on (x_j, η_{Bj}) it has mean $\sum_b \psi_b p_b(x_j, \eta_{Bj})$ and variance depending on $p_b(x_j, \eta_{Bj})$.

Given (τ_j, S_j), the law of (x_j, η_{Bj}) is fully determined except for σ^2_{uB}, via (B.8). Hence the conditional distribution

$$s_{Bj} \mid (\tau_j, S_j)$$

depends on σ^2_{uB} only through the variance term $\sigma^2_{x|\tau,S}(\tau_j, S_j) + \sigma^2_{uB}$ in (B.8). Variation in (τ_j, S_j) across proposals generates variation in both $\mu_{x|\tau,S}$ and $\sigma^2_{x|\tau,S}$, which in turn shifts the mean and dispersion of $s_{Bj} \mid (\tau_j, S_j)$ in a way that is strictly monotone in σ^2_{uB} under the Gaussian index structure. Therefore the mapping

$$\sigma^2_{uB} \longmapsto F_{s_B|\tau,S}(\cdot \mid t, \ell; \sigma^2_{uB}, \{\sigma^2_{\varepsilon_b}, k_b\}_b)$$

is one-to-one for all (t, ℓ) in the support of (τ_j, S_j), and the joint distribution of s_{Bj} given (τ_j, S_j) identifies σ^2_{uB}.

3. **Identification of** $(\sigma^2_{\varepsilon_b}, k_b)$ **from** $V_{bj} \mid (\tau_j, S_j)$**.** Fix blockholder b and treat (τ_j, S_j) as observable covariates. Conditional on (τ_j, S_j), we know:

   - the distribution of x_j;
   - the distribution of η_{Bj} via (B.8), with σ^2_{uB} now identified from Step 2;
   - the law of $z_{bj} = x_j + \varepsilon_{bj}$ with variance $\sigma^2_{\varepsilon_b}$.

   By linear-Gaussian updating, the posterior mean $m_{bj} = \mathbb{E}[x_j \mid \eta_{Bj}, z_{bj}]$ is a linear index in (x_j, η_{Bj}), and conditional on (τ_j, S_j) it is normally distributed with mean and variance that are known functions of $(\sigma^2_{\varepsilon_b}, k_b)$ and the known quantities $\mu_{x|\tau,S}(\tau_j, S_j)$ and $\sigma^2_{x|\tau,S}(\tau_j, S_j)$. Thus

$$\Pr\left(V_{bj} = 1 \mid \tau_j, S_j\right) = \Pr\left(m_{bj} \geq k_b \mid \tau_j, S_j\right) = \Phi\left(a_b(\tau_j, S_j; \sigma^2_{\varepsilon_b}, k_b)\right),$$

   for some smooth function $a_b(\cdot)$ that is strictly monotone in the underlying index. Observing the conditional choice probabilities (equivalently, the distribution of V_{bj}) over a rich support of (τ_j, S_j) therefore yields a standard probit-type identification problem in which the slope and intercept of the index are pinned down by cross-sectional variation in $\mu_{x|\tau,S}(\tau_j, S_j)$ and $\sigma^2_{x|\tau,S}(\tau_j, S_j)$. Under the normalization $x_j \sim N(0, 1)$, this uniquely identifies

$(\sigma^2_{\varepsilon_b}, k_b)$.

4. **Identification of δ_b.** Given $(\sigma^2_{uB}, \sigma^2_{\varepsilon_b}, k_b)$ and the already identified dispersed-side parameters, the equilibrium probability of being pivotal piv_{bj} as a function of the state x_j is known, and so is the posterior density $f(x_j \mid z_{bj} = k_b, \mathrm{piv}_{bj})$ in (15). The best-response condition (14),

$$\mathbb{E}\left[x_j \middle| z_{bj} = k_b,\ \mathrm{piv}_{bj}\right] = -\delta_b,$$

therefore uniquely determines δ_b.

These steps yield identification of σ^2_{uB} and $(\sigma^2_{\varepsilon_b}, k_b, \delta_b)$ for each blockholder b.

■

Lemma B.5 (Identification of preference parameters). *Suppose the primitives of the information environment and equilibrium cutoffs are identified as in Lemmas B.1 to B.4. In particular, assume that the econometrician knows:*

- *the dispersed information parameters $(\sigma^2_{uD}, \sigma^2_{\varepsilon D})$ and cutoff k_D;*
- *for each blockholder b, the public and private information parameters $(\sigma^2_{uB}, \sigma^2_{\varepsilon_b})$ and cutoff k_b;*
- *the proxy–advisor alignment parameter ρ and threshold ξ;*
- *the voting weights and pass threshold $(\{\psi_b\}_b, \Psi_D, \lambda^*)$ that determine pivotality.*

Then the dispersed preference parameter δ_D and each blockholder preference parameter δ_b are identified from the equilibrium best-response (indifference) conditions at the cutoffs:

$$\mathrm{E}\left[x_j \mid m_{Dj} = k_D,\ \mathrm{piv}_{Dj}\right] = -\,\delta_D, \tag{B.9}$$

$$\mathrm{E}\left[x_j \mid m_{bj} = k_b,\ \mathrm{piv}_{bj}\right] = -\,\delta_b, \qquad b = 1, \dots, N_B, \tag{B.10}$$

where m_{Dj} and m_{bj} denote the dispersed and blockholder posterior means defined in Section 2, and $\mathrm{piv}_{Dj}, \mathrm{piv}_{bj}$ are the corresponding pivotality events.

Proof. We give the argument for δ_D; the blockholder case is analogous.

1. **Posterior mean distribution at the cutoff.** Under the Gaussian information structure, the dispersed posterior mean $m_{Dj} = \mathrm{E}[x_j \mid \eta_{Dj}, z_{Dj}]$ is a linear function of $(x_j, u_{Dj}, \varepsilon_{Dj})$. In particular, conditional on $x_j = x$ it is normal:

$$m_{Dj} \mid x_j = x \sim N\left(\mu_{m_D}(x), \sigma^2_{m_D}\right),$$

where the mean $\mu_{m_D}(x)$ and variance $\sigma^2_{m_D}$ are explicit functions of $(\sigma^2_{uD}, \sigma^2_{\varepsilon D})$:

$$\mu_{m_D}(x) = \frac{\frac{1}{\sigma^2_{uD}} + \frac{1}{\sigma^2_{\varepsilon D}}}{\Lambda_D}\, x, \qquad \sigma^2_{m_D} = \frac{\Lambda_D - 1}{\Lambda_D^2}, \qquad \Lambda_D = 1 + \frac{1}{\sigma^2_{uD}} + \frac{1}{\sigma^2_{\varepsilon D}}.$$

Thus the conditional density of m_{Dj} at the cutoff k_D is known:

$$f_{m_D|x}(k_D \mid x) = \frac{1}{\sigma_{m_D}}\, \phi\left(\frac{k_D - \mu_{m_D}(x)}{\sigma_{m_D}}\right),$$

where ϕ is the standard normal PDF and all parameters are identified from the first-step information estimates.

2. **Pivotality as a function of the state.** Given the identified information environment and cutoffs $(k_D, \{k_b\}_b)$, the voting strategies of all other shareholders are known. For any state realization x, this pins down the distribution of the total vote from all other dispersed shareholders and blockholders, and therefore the probability that a representative dispersed shareholder is pivotal at the cutoff. Let

$$\pi_D(x) \equiv \Pr\left(\mathrm{piv}_{Dj} = 1 \mid x_j = x,\ m_{Dj} = k_D\right)$$

denote this pivotality kernel. It is a known function of the identified primitives and can be computed (analytically or numerically) from the equilibrium voting game.

3. **Bayes rule at the cutoff.** With the prior $x_j \sim N(0, 1)$ (density $\phi(x)$), Bayes rule implies

that the density of x_j conditional on $(m_{Dj} = k_D, \mathrm{piv}_{Dj})$ is proportional to

$$\phi(x)\, f_{m_D|x}(k_D \mid x)\, \pi_D(x).$$

Hence

$$\mathrm{E}\big[x_j \mid m_{Dj} = k_D, \mathrm{piv}_{Dj}\big] = \frac{\displaystyle\int_{-\infty}^{\infty} x\ \phi(x)\, f_{m_D|x}(k_D \mid x)\, \pi_D(x)\, dx}{\displaystyle\int_{-\infty}^{\infty} \phi(x)\, f_{m_D|x}(k_D \mid x)\, \pi_D(x)\, dx}.$$

Every object under the integrals is an explicit function of the identified parameters $(\sigma^2_{uD}, \sigma^2_{\varepsilon D}, k_D)$ and the equilibrium pivotality kernel $\pi_D(x)$, itself determined by $(\sigma^2_{uB}, \{\sigma^2_{\varepsilon_b}, k_b\}_b, \rho, \xi, \{\psi_b\}_b, \Psi_D, \lambda^*)$. Combining this with the best-response condition (B.9) gives the identification formula

$$\delta_D = -\frac{\displaystyle\int_{-\infty}^{\infty} x\ \phi(x)\, f_{m_D|x}(k_D \mid x)\, \pi_D(x)\, dx}{\displaystyle\int_{-\infty}^{\infty} \phi(x)\, f_{m_D|x}(k_D \mid x)\, \pi_D(x)\, dx},$$

which is a known scalar function of the previously identified primitives. Thus δ_D is identified.

4. **Blockholder preferences.** For each blockholder b, the same logic applies. Given $(\sigma^2_{uB}, \sigma^2_{\varepsilon_b}, k_b)$, the posterior mean m_{bj} satisfies

$$m_{bj} \mid x_j = x \sim N\big(\mu_{m_b}(x), \sigma^2_{m_b}\big),$$

with coefficients determined by $(\sigma^2_{uB}, \sigma^2_{\varepsilon_b})$. Let $f_{m_b|x}(k_b \mid x)$ denote the corresponding density at the cutoff. Define

$$\pi_b(x) \equiv \Pr\left(\mathrm{piv}_{bj} = 1 \mid x_j = x,\ m_{bj} = k_b\right),$$

the equilibrium pivotality probability for blockholder b, which is known from the identified information environment, cutoffs, and voting weights. Bayes rule then yields

$$\mathrm{E}\big[x_j \mid m_{bj} = k_b, \mathrm{piv}_{bj}\big] = \frac{\displaystyle\int x \; \phi(x)\, f_{m_b|x}(k_b \mid x)\, \pi_b(x)\, dx}{\displaystyle\int \phi(x)\, f_{m_b|x}(k_b \mid x)\, \pi_b(x)\, dx},$$

and the indifference condition (B.10) implies

$$\delta_b = -\frac{\displaystyle\int x \; \phi(x)\, f_{m_b|x}(k_b \mid x)\, \pi_b(x)\, dx}{\displaystyle\int \phi(x)\, f_{m_b|x}(k_b \mid x)\, \pi_b(x)\, dx}.$$

Therefore each δ_b is uniquely determined by the identified primitives.

Thus, given the identified information and cutoff parameters and the proxy-advisor alignment (ρ, ξ), the preference parameters $(\delta_D, \{\delta_b\}_b)$ are identified.

■

B.2. Non-Identification from Individual Voting Patterns

Proof of Corollary 1. The proof proceeds in three steps. Fundamentally, with only marginal data, we have $2 + 2N_B$ moments and $3 + 2n_B$ parameters, as the public information parameter σ_u^2 is not pinned down. We show that this non-identification filters through the information and preference environments, to deliver the same observed individual voting patterns.

1. **Observable content of the marginal distributions.** Write $\pi_i = 1/\sigma_{\varepsilon_i}^2$ for private-signal precision, $\pi_u = 1/\sigma_u^2$ for public-signal precision, and $\Lambda_i = 1+\pi_u+\pi_i$ for total posterior precision. In the large-N_D limit, the dispersed support rate is $\tau_j = \Phi\big(\sigma_{\varepsilon_D}^{-1}[(\Lambda_D - 1)x_j + \pi_u u_j - k_D\Lambda_D]\big)$. Define the probit-transformed support rate $T_j \equiv \Phi^{-1}(\tau_j)$ and the signal aggregate $Q_{Dj} \equiv (\Lambda_D - 1)x_j + \pi_u u_j$, so that $T_j = \sigma_{\varepsilon_D}^{-1}[Q_{Dj} - k_D\Lambda_D]$ with $Q_{Dj} \sim \mathcal{N}(0, \Sigma_D)$ and $\Sigma_D = \mathrm{Var}(Q_{Dj})$. The marginal distribution of τ_j is therefore pinned down by two quantities:

$$\mu_\tau \equiv \mathrm{E}[T_j] = -\frac{k_D\Lambda_D}{\sigma_{\varepsilon_D}}, \tag{B.11}$$

$$V_\tau \equiv \mathrm{Var}(T_j) = \frac{(\Lambda_D - 1)^2 + \pi_u}{\sigma_{\varepsilon_D}^2}. \tag{B.12}$$

   Similarly, define $Q_{bj} \equiv (\Lambda_b - 1)x_j + \pi_u u_j$ and let $\rho_{bD} = \mathrm{Cov}(Q_{Dj}, Q_{bj})/\Sigma_D$ and $\omega_{bD}^2 = \mathrm{Var}(Q_{bj}) - \mathrm{Cov}(Q_{Dj}, Q_{bj})^2/\Sigma_D$. Regressing Q_{bj} on Q_{Dj} and adding independent private noise gives the conditional blockholder voting probability

$$\Pr(V_{bj} = 1 \mid \tau_j) = \Phi\big(a_b \cdot T_j + c_b\big), \tag{B.13}$$

   where

$$a_b = \frac{\rho_{bD}\,\sigma_{\varepsilon_D}}{\sqrt{\omega_{bD}^2 + \pi_b}}, \qquad c_b = \frac{\rho_{bD}\,k_D\Lambda_D - k_b\Lambda_b}{\sqrt{\omega_{bD}^2 + \pi_b}}. \tag{B.14}$$

   Thus the marginal distribution of (τ_j, V_{bj}) for each blockholder b is fully characterized by $2 + 2N_B$ quantities: $(\mu_\tau, V_\tau, \{a_b, c_b\}_{b=1}^{N_B})$.

2. **Construction of an observationally equivalent parameter vector.** The deep parameters $(\sigma_u^2, \sigma_{\varepsilon_D}^2, k_D, \{\sigma_{\varepsilon_b}^2, k_b\}_{b=1}^{N_B})$ total $3 + 2N_B$ unknowns. We construct an alternative parameter vector that reproduces each marginal observable.

Fix $\tilde{\sigma}_u^2$ near σ_u^2, with $\tilde{\pi}_u = 1/\tilde{\sigma}_u^2$. Choose $\tilde{\sigma}_{\varepsilon_D}^2$ to match V_τ: from (B.12), this requires

$$\frac{(\tilde{\Lambda}_D - 1)^2 + \tilde{\pi}_u}{\tilde{\sigma}_{\varepsilon_D}^2} = V_\tau,$$

which pins down $\tilde{\sigma}_{\varepsilon_D}^2$ uniquely, and the solution exists by continuity for $\tilde{\sigma}_u^2$ near σ_u^2. Next, choose $\tilde{k}_D$ to match μ_τ: from (B.11),

$$-\frac{\tilde{k}_D \tilde{\Lambda}_D}{\tilde{\sigma}_{\varepsilon_D}} = \mu_\tau \qquad \Longrightarrow \qquad \tilde{k}_D = -\frac{\mu_\tau\, \tilde{\sigma}_{\varepsilon_D}}{\tilde{\Lambda}_D}.$$

For each blockholder b, choose $\tilde{\sigma}_{\varepsilon_b}^2$ to match a_b: from (B.14),

$$\frac{\tilde{\rho}_{bD}\, \tilde{\sigma}_{\varepsilon_D}}{\sqrt{\tilde{\omega}_{bD}^2 + \tilde{\pi}_b}} = a_b,$$

which is one equation in one unknown ($\tilde{\sigma}_{\varepsilon_b}^2$), with a solution by continuity. Finally, choose $\tilde{k}_b$ to match c_b: from (B.14),

$$\tilde{k}_b = \frac{\tilde{\rho}_{bD}\, \tilde{k}_D \tilde{\Lambda}_D - c_b \sqrt{\tilde{\omega}_{bD}^2 + \tilde{\pi}_b}}{\tilde{\Lambda}_b}.$$

3. **Equilibrium verification.** We verify that $(\tilde{k}_D, \{\tilde{k}_b\})$ are best responses under some preference profile. Define

$$\tilde{\delta}_D = -\mathrm{E}[x_j \mid \eta_j,\, \tilde{m}_{Dj} = \tilde{k}_D,\, \tilde{\mathrm{piv}}_{Dj}], \qquad \tilde{\delta}_b = -\mathrm{E}[x_j \mid \eta_j,\, \tilde{m}_{bj} = \tilde{k}_b,\, \tilde{\mathrm{piv}}_{bj}].$$

Intuitively, we are selecting the preference parameters to deliver the same observed cutoff profile under the new information environment, which works because the indifference conditions are well-defined. By construction, $\tilde{\Gamma}$ supports a BNE with cutoff profile $(\tilde{k}_D, \{\tilde{k}_b\})$ generating identical marginal distributions of (τ_j, V_{bj}) for each blockholder b.

■

B.3. Estimation Strategy

This section formalizes the estimation routine.

- **Analytic moments.** We derive closed-form expressions for all model-implied moments used in estimation (including ISS–dispersed moments and blockholder voting moments), so the moment mapping $g(\theta)$ is computed analytically and is fully consistent with the model's equilibrium voting rules.

- **Single GMM objective** Although it is helpful to describe the parameters as belonging to distinct components of the model (e.g., ISS/dispersed information, blockholder behavior, and preference parameters), we estimate the full parameter vector jointly in a single GMM objective. All moments enter one stacked moment vector $\hat{g}(\theta)$,

$$\hat{\theta} = \arg\min_{\theta} \ \hat{g}(\theta)^{\top} \hat{W} \, \hat{g}(\theta),$$

  is our GMM objective.

- **Two-step GMM.** We implement a standard two-step routine: a first step using an initial weighting matrix (e.g., identity) to obtain $\hat{\theta}^{(1)}$, followed by a second step using the efficient weighting matrix $\hat{W} = \widehat{\Omega}(\hat{\theta}^{(1)})^{-1}$, where $\widehat{\Omega}$ is the estimated covariance matrix of the sample moments.

- **Blockholder probit moments as GMM restrictions.** Blockholder probit-style moments are implemented as equivalent GMM moment restrictions: we include the orthogonality conditions implied by the probit specification.

- **Inference.** Standard errors follow conventional GMM asymptotics, using the estimated Jacobian of moments and the estimated moment covariance matrix (sandwich form) evaluated at $\hat{\theta}$.